A profinite reduction of Erdős Problem 25

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Abstract

Let $1 \le n_1 < n_2 < \cdots$ be integers with residues a_i, and let A be the set of integers n such that for every i either $n < n_i$ or $n \not\equiv a_i \pmod{n_i}$. Erdős asked whether A must have a logarithmic density. We recast the question in the profinite completion $\widehat{\mathbb{Z}}$, where each class is a ball of Haar measure $1/n_i$ and the survivors have a Haar measure ν that always exists, and prove unconditionally that the upper logarithmic density of A is at most ν, so that the reverse inequality would settle the problem. We prove the reverse inequality in a series of regimes: convergent drift; pairwise incompatible, pairwise coprime, aligned and boundedly misaligned systems; systems admitting a cylinder decomposition; and, through reduction theorems – exhaustion of dying cylinders, confinement to protectors of summable measure, and a theorem for coprime-layered trees of cylinders proved by a martingale argument – systems of divergent drift, positive survivor measure and unbounded misalignment whose classes are organised along nested protectors, pairwise coprime protectors, or protectors that are the pairwise products of a set of primes, each under explicit and restrictive hypotheses on the layers. A second-moment condition, with a lemma restoring the zero–one law for families whose indices do not escape to infinity, gives a sufficient condition for $\nu = 0$. We exhibit the step of the Davenport–Erdős argument that shifted residues break, refute four natural routes by explicit construction, bound a fifth, and show that every system is majorised by a saturated one whose classes are the maximal cylinders meeting its survivor set in a null set. What remains is stated as a checklist that a counterexample must pass in full and, without congruences, as a question about a closed subset of $\widehat{\mathbb{Z}}$ of positive measure whose maximal holes have divergent total measure. The problem is not resolved here.

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1 Introduction

Let $1 \le n_1 < n_2 < \cdots$ be a strictly increasing sequence of integers and let $a_i \in \mathbb{Z}$ be arbitrary. The ith class kills the set

$$K_i = \{x \ge n_i : \ x \equiv a_i \pmod{n_i}\},$$

and the survivor set is $A = \mathbb{N} \setminus \bigcup_i K_i$. We write $\delta(E)$ for the natural density of a set E when it exists, and $\delta_{\log}(E)$ for its logarithmic density. Erdős asked whether A must possess a logarithmic density,

$$\delta_{\log}(A) = \lim_{X\to\infty} \frac{1}{\log X} \sum_{m\in A,\ m<X} \frac{1}{m}.$$

The truncation $x \geq n_i$ is essential: without it each class is a full residue class and the question becomes a statement about sets of multiples only in the aligned case.

When every $a_i \equiv 0$ the killed set is a set of multiples and the answer is affirmative by the theorem of Davenport and Erdős [2, 3]; see [17] for a modern account. The general problem has resisted precisely because shifting the residues destroys the divisibility structure that their proof consumes. Our purpose is to say exactly what that destruction costs, to prove as much of the statement as survives it, and to isolate what remains.

We do not claim the individual ingredients are deep. The profinite upper bound is elementary; the regime results rest on the Chinese remainder theorem, Davenport–Erdős and a standard Tauberian argument. What we offer is an accounting: the step of the classical proof that misalignment breaks, a weighted form of Davenport–Erdős that survives it, the regimes and reductions under which the reverse inequality is recovered, a single uniform statement whose truth would settle the rest, and four natural approaches shown by explicit construction not to work, together with a bound on a fifth, the most natural attempt at a counterexample.

Results

Sections 2 and 3 set up the profinite framework and prove the unconditional half of the problem (Theorem 3.1), together with the threshold convention and the restriction lemma by which systems pass to cylinders. Section 4 records two lemmas on sets of multiples (Lemmas 4.1 and 4.2), the second a weighted form of Davenport–Erdős. Section 5 proves the reverse inequality for convergent drift, incompatible systems, translated systems and finitely many aligned channels (Theorems 5.1, 5.3, 5.9, 5.11), proves $\nu = 0$ under coprime divergence (Theorem 5.4), and gives the second-moment criterion for $\nu = 0$ (Theorem 5.5, Lemma 5.6). Section 6 exhibits the step of the Davenport–Erdős argument that shifted residues break (Proposition 6.1). Section 7 proves what survives it: bounded misalignment and cylinder decompositions (Theorems 7.2, 7.5, 7.6). Section 8 contains the reduction theorems: exhaustion and saturation (Theorems 8.1, 8.4), summable protectors (Theorem 8.8), coprime-layered trees (Theorem 8.12), and coprime and pairwise-product protectors (Theorems 8.16, 8.18). Section 9 gives a uniform statement that suffices for the rest (Theorem 9.1), its excursion form, and the checklist and closed-set question in which we leave the problem. Section 10 refutes four routes by construction and bounds a fifth; Section A describes the verification. We emphasise at the outset that the problem is *not* resolved here.

The problem in the literature. The question is Problem 25 on the Erdős problems site [4], listed as open, and as a special case of Problem 486, in which several residues may be excluded modulo each n; its statement has been formalised in Lean. Tenenbaum has recorded that Erdős was much interested in whether the Davenport–Erdős theorem extends to non-zero residues, and that he himself has thought about it without finding a promising starting point [8]. A neighbouring question, Problem 281, asks: if $n_1 < n_2 < \cdots$ is such that for every choice of residues the integers avoiding all the classes have density 0, must there be, for every $\varepsilon > 0$, a k such that for every choice of residues the integers avoiding the first k classes have density below ε? It was answered affirmatively in 2026 by the observation (recorded in [7]) that an old theorem of Rogers (Halberstam–Roth [6], Ch. V) applies: given ε, the 1936 Davenport–Erdős theorem gives a k for which the *aligned* survivors of the first k classes have density below ε, and Rogers' inequality then bounds the survivors of the same k moduli with any residues by that aligned density. It does not bear on the existence of the density. The only prior progress on Problem 25 itself that we know of is a 2026 note of Chojecki [5], which proves that the natural density exists when the drift converges (his Theorem 3.1; Theorem 5.1 below is its logarithmic form) and when all the moduli are pairwise coprime (his Theorem 3.2; for divergent drift this is the case of Theorem 5.4 below in which the whole family is coprime), and reduces the general problem to a uniform harmonic estimate for the finite “quotient sieves” by which each class is first reached, proving two obstructions: the tail union admits no vanishing logarithmic bound, and a prime-power tower gadget gives a single first-kill set a transient harmonic contribution far larger than its eventual density loss. In the language of this paper the first is the failure of the union bound under divergent drift; the second is the transient-spike phenomenon whose aggregate effect on the logarithmic average Proposition 9.4 bounds. Apart from those two theorems we have found no prior source for the results below; the weighted Davenport–Erdős lemma in particular is an elementary variant of the classical Dirichlet-series proof and we claim no more for it than its use here. We note also where the neighbouring literature stops. The lower bounds on the density of integers left uncovered by a system of congruences – the inequality of Filaseta, Ford, Konyagin, Pomerance and Yu [13], a product term less a pairwise-dependence correction, and Hough’s relative form of the local lemma [14] – concern the density of the periodic set uncovered by finitely many classes, that is the profinite measure ν of the present paper. They bound ν from below; the open half of Problem 25 is a lower bound on the integer survivors at scales far below the period, and no estimate of that kind appears in this literature. Erdős’s own criterion [12] for a set of multiples to have a natural density – that integers whose least divisor in the set lies in $(n^{1-\varepsilon}, n]$ be rare – concerns the same top-of-range phenomenon of which Proposition 9.4 controls only the single dyadic window $(X/2, X]$, and only in logarithmic density; the range $(X^{1-\varepsilon}, X/2]$ is exactly what remains uncontrolled here. The theory of Behrend sequences – sets of multiples of full density – is the aligned case’s version of the question when $\nu = 0$, and it is instructive how hard even that is. Hall and Tenenbaum [15] and Tenenbaum [16] obtain sharp criteria only for block sequences, in the form of Borel–Cantelli conditions whose exponent

$$\alpha = \frac{1 - \log 2}{1 - (1 + \log\log 2)/\log 2} \approx 3.565,$$

in Tenenbaum’s words, measures the level of mutual dependence among the events ([16], the display following (1.7)); the method conditions on the product n_k of an integer’s prime factors below $\exp\exp k$ and iterates bounds on the conditional probability of a new divisor, which is the multiplicative counterpart of the cylinder filtration in Theorem 8.12. Tenenbaum describes effective general criteria as very difficult, if not hopeless. For shifted classes, sufficient conditions for $\nu = 0$ are given here by the cruder second-moment route of Theorems 5.4 and 5.5 and Lemma 5.6; the open half of Problem 25 concerns $\nu > 0$, a case which for sets of multiples is entirely Davenport–Erdős’s and has no counterpart in that literature.

2 The profinite framework

Write $\widehat{\mathbb{Z}} = \varprojlim \mathbb{Z}/m\mathbb{Z}$ for the profinite integers with normalised Haar measure μ.

Definition 2.1. To the class (n, a) associate the ball $B(n, a) = a + n\widehat{\mathbb{Z}} \subseteq \widehat{\mathbb{Z}}$, which is clopen with $\mu(B(n, a)) = 1/n$. Write $B_i = B(n_i, a_i)$ and

$$\widehat{U} = \bigcup_i B_i, \qquad \widehat{A} = \widehat{\mathbb{Z}} \setminus \widehat{U}, \qquad \nu = \mu(\widehat{A}).$$

Throughout, a subscript on ν names the subsystem whose survivor measure is meant – the first N classes (ν_N), the classes of modulus at most X (ν_X), a set of classes or groups (ν_T, ν_J) – and a prime marks a reduced system; for a cylinder, ν_c or ν_C is the survivor measure of the induced system there, that is, the measure relative to the cylinder. We fix once and for all an enumeration of the classes by non-decreasing modulus, ties in any fixed order; "the first N classes" refers to it. Nothing below depends on the choice: finite heads enter only through limits over N, and the same quantities can be defined by a modulus cutoff instead. A system is a *set* of classes: no ball is listed twice, a repetition killing nothing new. Several classes may share a modulus, a slight widening of the problem in which Erdős's case is that of strictly increasing moduli; the reductions of Section 8 produce systems of this wider kind. Since a modulus n carries at most n residues, a system has at most n classes of modulus n, so bounded modulus still means finitely many classes, and an infinite system has unbounded moduli. We reserve the hat $\widehat{U}$ for this subset of $\widehat{\mathbb{Z}}$ and write $U = \bigcup_i K_i \subseteq \mathbb{N}$ for the set of killed integers; likewise $\widehat{A} \subseteq \widehat{\mathbb{Z}}$ is the profinite survivor set and $A \subseteq \mathbb{N}$ the integer one. The profinite and integer objects are never identified.

Since $\widehat{U}$ is open, $\mu(\widehat{U})$ exists, and therefore ν *always exists*, with no hypothesis on the system. This is the first gain of the profinite viewpoint: the candidate value of the answer is available before anything has been proved.

Lemma 2.2. *Two classes (n, a) and (n', a') have a common solution if and only if $a \equiv a' \pmod{\gcd(n, n')}$, and this holds if and only if $B(n, a) \cap B(n', a') \neq \emptyset$. A family of classes is pairwise compatible if and only if it has a simultaneous solution* in $\widehat{\mathbb{Z}}$, *equivalently if and only if the corresponding balls have a common point. For infinite families that solution need not be an integer; see Remark 2.4.*

Proof. The first assertion is the two-modulus case of the Chinese remainder theorem. For the second, note that $\widehat{\mathbb{Z}} \cong \prod_p \mathbb{Z}_p$ and a ball is the product of its p-adic components. In $\mathbb{Z}_p$ balls are ultrametric: if two intersect then one contains the other. Hence any finite pairwise-intersecting family has a ball of smallest radius, which is contained in all the others, so the family has the finite intersection property; compactness of $\mathbb{Z}_p$ then gives a nonempty intersection in each coordinate. $\square$

Definition 2.3. A *channel* is a pairwise compatible family of classes; by Lemma 2.2 this is exactly a family of balls sharing a common point of $\widehat{\mathbb{Z}}$. A channel is *aligned* if that common point may be taken in $\mathbb{Z}$, i.e. if there is an integer c with $a_i \equiv c \pmod{n_i}$ for all its members. The *drift* of a family is $\sum 1/n_i$.

Remark 2.4. The two notions coincide for finite families, where the Chinese remainder theorem produces an integer solution modulo the lcm, but not in general. An infinite channel has a common point of $\widehat{\mathbb{Z}}$, and $\widehat{\mathbb{Z}}$ is uncountable while $\mathbb{Z}$ is countable, so a coherent sequence of residues modulo 2^k may be chosen avoiding every integer; the resulting family is pairwise compatible but aligned to no integer. The distinction is not cosmetic: the translation argument of Theorem 5.9 requires an integer, so it applies to aligned channels only.

3 The unconditional half

Theorem 3.1. *For every system, $\overline{\delta}_{\log}(A) \leq \nu$. No hypothesis on the moduli or residues is required.*

Proof. Fix N and let A_N be the survivor set of the first N classes. The truncated condition $x \equiv a_i$ and $x \geq n_i$ differs from the profinite condition $x \in B_i$ only at $x = a_i \bmod n_i$ when that representative is smaller than n_i, so the two survivor sets of the first N classes differ by at most N integers. A finite set has density zero, and the profinite survivor set of finitely many classes is a finite union of residue classes, hence periodic; so A_N has a natural density and a logarithmic density, both equal to $\nu_N := \mu(\widehat{\mathbb{Z}} \setminus \bigcup_{i \leq N} B_i)$. Since $A \subseteq A_N$ we get $\overline{\delta}_{\log}(A) \leq \nu_N$ for every N, and $\nu_N \downarrow \nu$ by continuity of measure from above. □

Corollary 3.2. *If* $\nu = 0$ *then* $\delta_{\log}(A)$ *exists and equals* 0.

Proof. By Theorem 3.1, $\overline{\delta}_{\log}(A) \leq \nu = 0$, and logarithmic densities are non-negative. □

Thresholds. Passing to a cylinder, as we shall do repeatedly, changes the point at which a class begins to kill. A *threshold system* consists of classes $(n_i, a_i; t_i)$, the class killing the integers $x \geq t_i$ with $x \equiv a_i \pmod{n_i}$, where $1 \leq t_i \leq n_i$ and, for some constant M' and all but finitely many i, $t_i \geq n_i/M'$. The systems of the introduction are the case $t_i = n_i$. Thresholds are invisible to every profinite quantity, since balls do not see them. On the integer side they enter in exactly five ways. A finite head remains periodic up to finitely many integers. The harmonic sum of a class over $[1, X]$ is at most $(\log X + 1)/n_i + 1/t_i \leq (\log X + 1 + M')/n_i$, so every tail bound of the form $\sum_{\text{tail}}(\log X + O(1))/n_i$ keeps its shape with a larger constant. And a class whose induced condition is a pure divisibility $m \mid y$ is unchanged by a threshold at most m, since the only multiple of m below m is 0. Finally, a class of modulus n lying inside a cylinder of modulus d kills only integers $x \geq t_i \geq n/M' \geq d/M'$, so its kills lie in $C \cap [d/M', \infty)$, whose harmonic sum up to X is at most $(\log X + \log M' + M')/d$: the tail bounds of Section 8, which sum such quantities over cylinders, keep their shape with a larger constant. There is one further point of contact, on the side of natural rather than logarithmic density: a class of modulus n_i kills at most $X/n_i + 1$ integers below X, the term 1 being absent for a standard class, and only classes with $t_i \leq X$, hence $n_i \leq M'X$, contribute; when $\sum_i 1/n_i < \infty$ the number of such classes is $o(X)$, since $\#\{i : Y/2 < n_i \leq Y\} \leq Y \sum_{n_i > Y/2} 1/n_i = o(Y)$ summed over dyadic ranges, so the accumulated extra terms are $o(X)$. Every result of this paper is stated for standard systems and holds for threshold systems with the same proof, thresholds entering only through these five facts: Theorems 3.1 and 5.4 through the first, Theorem 5.1 and Lemma 4.1 through the second, Theorem 7.2 through the third, Theorems 8.1, 8.8 and Lemma 8.11 through the fourth, with the system's constant M' in place of 1, and Theorem 5.3 through the fifth – with one exception, Theorem 8.4, which is stated for standard systems and is false for threshold systems: the classes $(2^k, 2^{k-1}; 2^{k-1})$, $k \geq 1$, kill every positive integer, while their balls fill $2\widehat{\mathbb{Z}}$ up to the point 0, so $\nu = \frac{1}{2}$ and the saturated system $\{2\widehat{\mathbb{Z}}\}$ has integer survivors of logarithmic density $\frac{1}{2}$. Its proof uses that a class kills nothing below its modulus. The theorem is applied only to the original system. Induced systems arise in two ways only. Theorems 7.2, 7.5 and 7.6 restrict to the cylinders of one fixed modulus M, and Lemma 3.3 gives those induced systems the constant $2MM'$; the tree theorem restricts to the finitely many cells of one fixed depth, each with its own such constant. Theorem 8.1 also speaks of the induced system of each group on its own cylinder, of unbounded modulus, but only to apply Corollary 3.2 to each one separately, which needs no threshold constant at all. No argument restricts repeatedly with unbounded moduli: the reduction theorems of Section 8 act on the original system, and the fourth fact bounds their tail kills by the original thresholds. The correspondence between a system and an induced one holds up to the finitely many integers below the cylinder's modulus, which no density sees.

Lemma 3.3 (Restriction)**.** *Let $c + M\mathbb{Z}$ be a cylinder with $0 \le c < M$ and write $x = c + My$, $y \ge 1$. A class $(n, a; t)$ either misses the cylinder, when $\gcd(n, M) \nmid a - c$, or becomes in y the class of modulus $m = n/\gcd(n, M)$, with the residue determined by a and c, and threshold $t' = \max\bigl(1, \lceil (t-c)/M \rceil\bigr) \le m$. For all but finitely many classes $t' \ge m/(2MM')$, so the induced family is a threshold system, and, apart from the finitely many integers below M, the integers of the cylinder surviving the original system are exactly $c + My$ for y surviving the induced one.*

Proof. The congruence $c + My \equiv a \pmod{n}$ has a solution iff $\gcd(n, M) \mid a - c$, and then its solutions form one class modulo m. The condition $c + My \ge t$ is $y \ge (t - c)/M$, and $(t - c)/M \le n/M \le m$. Since $t \ge n/M'$ for all but finitely many classes, $t' \ge n/(MM') - 1 \ge m/(MM') - 1 \ge m/(2MM')$ once $m \ge 2MM'$, which excludes only the finitely many classes with $n \le 2M^2M'$ (finitely many, a system having at most n classes of modulus n). The last sentence is the definition of the induced system. □

Remark 3.4 (Rogers' theorem)**.** The measure ν is largest when all residues vanish. A theorem communicated by Rogers and published only in Halberstam and Roth [6, Ch. V] – see Tao's exposition [7] – states that in a finite cyclic group the union of cosets $a_j + H_j$ is at least as large as the union of the subgroups H_j. Applied in $\mathbb{Z}/L_N\mathbb{Z}$ to the first N classes and letting $N \to \infty$, it gives $\nu \le \nu_{\mathrm{al}}$, where ν_{al} is the profinite survivor measure of the aligned system, which by Davenport–Erdős is the logarithmic density of the integers divisible by no modulus. With Theorem 3.1 the chain $\overline{\delta}_{\log}(A) \le \nu \le \nu_{\mathrm{al}} = \delta_{\log}\{m : \ n_i \nmid m \ \forall i\}$ shows that the survivors of *any* residue choice have upper logarithmic density at most the logarithmic density of the non-multiples of the same moduli. The inequality runs the wrong way for the open half of the problem, which is a lower bound on the survivors.

So every system whose survivors vanish in measure is settled by the upper bound alone, with no further hypothesis; the open problem lives entirely in the range $\nu > 0$.

A sufficient form of Erdős's question is therefore the reverse inequality $\underline{\delta}_{\log}(A) \ge \nu$, which asserts that the classes beyond any finite head destroy no more than their own measure. It is sufficient and not equivalent: a system whose survivors had a logarithmic density strictly below ν would answer the question affirmatively while violating it, and nothing here excludes that. Every positive result below proves the inequality, and the reductions of Section 9 are stated for it.

4 Two lemmas on sets of multiples

We isolate two facts used in Sections 5–8. Write $M(D)$ for the set of positive multiples of members of D, and $D^{(N)}$ for the N smallest members of D.

Lemma 4.1 (Vanishing tails)**.** *Let D be a set of moduli and let $D_1 \subseteq D_2 \subseteq \cdots$ be any increasing exhaustion of D by finite sets. Then $\delta_{\log}\bigl(M(D) \setminus M(D_N)\bigr) \to 0$ as $N \to \infty$. In particular this holds for $D_N = D^{(N)}$, the N smallest members.*

Proof. Each $M(D_N)$ is a finite union of sets of multiples, hence periodic, so it has a natural density $\delta(M(D_N))$, and these increase with N because the sets do. The family (D_N) is cofinal in the directed family of finite subsets of D: every finite $F \subseteq D$ lies in some D_N, since F is finite and the D_N exhaust D. Therefore

$$\lim_N \delta\bigl(M(D_N)\bigr) \;=\; \sup_{F \subseteq D \text{ finite}} \delta\bigl(M(F)\bigr) \;=\; \delta_{\log}\bigl(M(D)\bigr),$$

the last equality being the theorem of Davenport and Erdős [2, 3] (see [17, Ch. 0]), which in particular gives that $M(D)$ has a logarithmic density. Note that the limit is thus the same for every exhaustion, not only for the one by the N smallest members.

The union

$$M(D) = M(D_N) \ \sqcup \ \big(M(D) \setminus M(D_N)\big)$$

is disjoint, and logarithmic density is finitely additive on disjoint sets whenever two of the three quantities exist; here $M(D)$ and $M(D_N)$ both have logarithmic densities, so the difference set has one, equal to $\delta_{\log}(M(D)) - \delta(M(D_N))$, which tends to 0. □

Lemma 4.2 (Weighted Davenport–Erdős). *Let $\varphi : \mathbb{N} \to [0,1]$ satisfy $\varphi(d) \le \varphi(n)$ whenever $d \mid n$. Then $\lim_{X\to\infty} \frac{1}{\log X} \sum_{n\le X} \varphi(n)/n$ exists.*

Proof. If $\varphi \equiv 0$ the assertion is trivial, so assume $\varphi(n_0) > 0$ for some n_0; then $F(\sigma) := \sum_{n\ge 1} \varphi(n) n^{-\sigma} > 0$ for $\sigma > 1$, and division by F below is legitimate. The series converges absolutely for $\sigma > 1$ since $\varphi \le 1$. Monotonicity gives

$$\sum_{d\mid n} \varphi(d)\Lambda(n/d) \ \le \ \varphi(n) \sum_{d\mid n} \Lambda(n/d) \ = \ \varphi(n)\log n,$$

the inequality that drives the Davenport–Erdős argument and which reappears as (1) below. Multiplying by $n^{-\sigma}$ and summing, the left side is the Dirichlet convolution $F(\sigma)\cdot(-\zeta'/\zeta)(\sigma)$ and the right side is $-F'(\sigma)$, so

$$-F'(\sigma) \ \ge \ F(\sigma)\Big(-\frac{\zeta'}{\zeta}\Big)(\sigma), \qquad \text{i.e.} \qquad \frac{d}{d\sigma}\log\frac{F(\sigma)}{\zeta(\sigma)} \ \le \ 0.$$

Hence F/ζ is non-increasing on $(1,\infty)$ and takes values in $[0,1]$, so $L = \lim_{\sigma\to 1^+} F(\sigma)/\zeta(\sigma)$ exists. Since $(\sigma-1)\zeta(\sigma) \to 1$ we get $(\sigma-1)F(\sigma) \to L$, and as the coefficients $\varphi(n)$ are non-negative, Karamata's Tauberian theorem for the Laplace–Stieltjes transform of the non-decreasing function $t \mapsto \sum_{\log n\le t} \varphi(n)/n$ (see [20, Ch. II.7]) converts $(\sigma-1)F(\sigma) \to L$ into $\sum_{n\le X} \varphi(n)/n \sim L\log X$ when $L > 0$, the stated logarithmic mean. When $L = 0$ the conclusion $\sum_{n\le X} \varphi(n)/n = o(\log X)$ follows from positivity alone: with $\alpha(t) = \sum_{\log n\le t} \varphi(n)/n$ one has $\alpha(t) \le e\,F(1+1/t)$, since every term with $\log n \le t$ carries the factor $n^{-1/t} \ge e^{-1}$ in $F(1+1/t)$, so $\alpha(t)/t \le e\,t^{-1}F(1+1/t) \to 0$. Note that nothing beyond $0 \le \varphi \le 1$ and divisibility-monotonicity was used; in particular φ need not be an indicator function, which is the case of Davenport–Erdős. □

5 Five regimes in which the reverse inequality holds

Theorem 5.1 (Convergent drift). *If $\sum_i 1/n_i < \infty$ then $\delta_{\log}(A)$ exists and equals ν, for every choice of residues.*

(This is Theorem 3.1 of Chojecki [5], who obtains the natural density; the proof below is included because its harmonic union bound is used throughout.)

Proof. We first record the harmonic union bound, which is what the logarithmic setting requires: a bound on natural density would not suffice, since a set of small natural density can carry large logarithmic mass.

The elements of K_i below X are at least n_i and lie in an arithmetic progression of modulus n_i, so the jth of them is at least $j\,n_i$; hence

$$\frac{1}{\log X}\sum_{x\in K_i,\ x\le X}\frac{1}{x}\ \le\ \frac{1}{n_i\log X}\sum_{j\le X/n_i}\frac{1}{j}\ \le\ \frac{1}{n_i}\Big(1+\frac{1}{\log X}\Big).$$

Summing and using $1_{\bigcup K_i}\le\sum 1_{K_i}$ pointwise,

$$\limsup_{X\to\infty}\frac{1}{\log X}\sum_{x\in\bigcup_{i>N}K_i,\ x\le X}\frac{1}{x}\ \le\ \sum_{i>N}\frac{1}{n_i}.$$

Now given $\varepsilon>0$ choose N with $\sum_{i>N}1/n_i<\varepsilon$. Then $A\subseteq A_N$ and $A_N\setminus A\subseteq\bigcup_{i>N}K_i$, whose upper logarithmic density is below ε by the display above. Since A_N is periodic apart from a finite symmetric difference with the profinite survivor set of the first N classes, $\delta_{\log}(A_N)=\nu_N$; hence the upper and lower logarithmic densities of A satisfy $\nu_N-\varepsilon\le\underline{\delta}_{\log}(A)\le\overline{\delta}_{\log}(A)\le\nu_N$, so they differ by at most ε. As ε was arbitrary they agree; the common value is at most ν by Theorem 3.1 and at least $\nu_N-\varepsilon\ge\nu-\varepsilon$ for every ε, so it is ν. □

Lemma 5.2 (Disjointness bound)**.** *If the classes are pairwise incompatible then* $\sum_i 1/n_i\le 1$, *and the constant* 1 *is attained.*

Proof. Incompatible classes have no common solution, so the sets K_i are pairwise disjoint; each has density $1/n_i$ since truncation removes finitely many elements. For any finite subfamily the disjoint union lies in $\mathbb{N}$, giving $\sum 1/n_i\le 1$; take the supremum. Attainment: the class $(2,0)$ together with $(12,r)$ for the six odd r is pairwise incompatible and has drift $\frac{1}{2}+6\cdot\frac{1}{12}=1$. (Before truncation these residue classes cover $\mathbb{Z}$; after truncation the odd integers below 12 survive, which does not affect the drift computation.) □

Theorem 5.3 (Incompatible systems)**.** *If the classes are pairwise incompatible then the natural density of* A *exists and equals* $1-\sum_i 1/n_i$*; in particular* $\delta_{\log}(A)$ *exists and equals* ν*.*

Proof. The K_i are pairwise disjoint, so for each N the density of $\bigcup_{i\le N}K_i$ is exactly $\sum_{i\le N}1/n_i$, giving $\underline{\delta}\big(\bigcup_i K_i\big)\ge\sum_{i\le N}1/n_i$ for every N. For the reverse bound, the elements of K_i below X are at least n_i and lie in a progression of modulus n_i, so there are at most X/n_i of them (at most X/n_i+1 for a threshold class, the extra terms summing to $o(X)$ by the fifth threshold fact); disjointness then gives $\big|\bigcup_i K_i\cap[1,X]\big|\le X\sum_i 1/n_i+o(X)$, whence $\overline{\delta}\le\sum_i 1/n_i$. By Lemma 5.2 that sum converges, so upper and lower densities agree. □

Theorem 5.4 (Coprime divergence)**.** *If some pairwise coprime subfamily has divergent drift then* A *has density* 0, *and* $\delta_{\log}(A)=0=\nu$*.*

(When the whole family is pairwise coprime this is contained in Theorem 3.2 of Chojecki [5].)

Proof. Coprime moduli give independent kill events by the Chinese remainder theorem, so the survivor density through the first N members of the subfamily is exactly $\prod(1-1/n_i)$, which tends to 0 when $\sum 1/n_i$ diverges. Above the largest of those N moduli, A is contained in each of these periodic sets, so $\delta(A)=0$. The same product tends to 0 in measure, so $\nu=0$. □

The coprime hypothesis can be replaced by a second-moment condition on the kills, grouped by smallest prime.

Theorem 5.5 (Quasi-independence)**.** *For each prime p let $\widehat{K}_p \subseteq \widehat{\mathbb{Z}}$ be the union of the balls of the classes whose modulus has smallest prime factor p. If $\sum_p \mu(\widehat{K}_p) = \infty$ and*

$$\limsup_{N\to\infty} \frac{\sum_{p,p'\le N} \mu(\widehat{K}_p \cap \widehat{K}_{p'})}{\left(\sum_{p\le N} \mu(\widehat{K}_p)\right)^2} < \infty,$$

then $\nu = 0$, and hence $\delta_{\log}(A) = 0$.

Proof. Haar measure on $\widehat{\mathbb{Z}} = \prod_\ell \mathbb{Z}_\ell$ is the product of the Haar measures of the coordinates, so the coordinates are independent. Every prime dividing a modulus with smallest prime factor p is at least p, so $\widehat{K}_p$ is determined by the coordinates at primes $\ell \ge p$. Hence $E = \{x : x \in \widehat{K}_p$ for infinitely many $p\}$ lies in the tail σ-algebra of the coordinates, and by Kolmogorov's zero–one law $\mu(E) \in \{0,1\}$. The Kochen–Stone lemma [10], the second-moment form of Borel–Cantelli, gives under the two hypotheses $\mu(E) \ge 1/\limsup_N(\cdots) > 0$. So $\mu(E) = 1$. Every point of E lies in a ball, so the survivor set in $\widehat{\mathbb{Z}}$ is contained in $\widehat{\mathbb{Z}} \setminus E$ and $\nu = 0$; Corollary 3.2 finishes. □

The hypothesis that $\widehat{K}_p$ be indexed by smallest primes is only a way of forcing the sets to move off to infinity in the coordinates. What the proof uses is the following, and it will be needed in a case where the index is not a prime at all.

Lemma 5.6 (Escaping families)**.** *Let $(\widehat{K}_j)$ be one countable family of measurable subsets of $\widehat{\mathbb{Z}}$, and for each P let $\mathcal{K}_P$ be the set of its members determined by the coordinates at primes exceeding P; these sets decrease as P grows. Suppose that for every P the members of $\mathcal{K}_P$ can be enumerated so that their measures have divergent sum and the second-moment ratios along some subsequence of that enumeration have limsup at most a constant ρ independent of P. Then almost every point of $\widehat{\mathbb{Z}}$ lies in infinitely many $\widehat{K}_j$.*

Proof. Fix P and let E_P be the set of points lying in infinitely many members of $\mathcal{K}_P$. Kochen–Stone [10] gives $\mu(E_P) \ge 1/\rho$. The families $\mathcal{K}_P$ decrease as P grows, so the E_P decrease, and $\mu(\bigcap_P E_P) = \lim_P \mu(E_P) \ge 1/\rho > 0$. Each E_P is determined by the coordinates above P, so the intersection lies in the tail σ-algebra and has measure 1 by Kolmogorov's zero–one law. It is contained in the set of points lying in infinitely many $\widehat{K}_j$. □

Remark 5.7. The distinction matters. In Theorem 5.5 the sets are indexed by their smallest prime and escape automatically. In Theorem 8.18 below they are indexed by pairs of primes, and infinitely many of them involve any given small prime, so the set of points covered infinitely often is *not* a tail event; only after discarding the pairs that touch the first $\pi(P)$ primes does the argument apply, and it is the limit over P that restores the zero–one law.

Remark 5.8. Theorem 5.4 is the case in which the subfamily's smallest primes are distinct and each $\widehat{K}_p$ is a single ball: then $\mu(\widehat{K}_p \cap \widehat{K}_{p'}) = \mu(\widehat{K}_p)\mu(\widehat{K}_{p'})$ by the Chinese remainder theorem and the ratio tends to 1. Read in contrapositive the theorem is a structure statement: a system with $\nu > 0$ has either $\sum_p \mu(\widehat{K}_p) < \infty$ – the kills through each smallest prime are summable, as when the smallest primes of the moduli are themselves sparse – or smallest-prime kills so positively correlated that the ratio is unbounded. The many-protector systems of Section 8 – systems whose classes fall into groups each confined to a cylinder, its *protector*; when the protector's modulus is a prime p_j we also call p_j the protector – fall on either side according to the sparsity of the protectors: a class $p_j m$ has smallest prime $\min(p_j, p(m))$, so $\widehat{K}_p$ collects only the protectors $p_j \geq p$ and $\sum_p \mu(\widehat{K}_p) \approx \sum_j \pi(p_j)/p_j$, which behaves like $\sum_j 1/\log p_j$: convergent for $p_j \geq \exp(j^2)$, say, and divergent for exponentially or polynomially sparse p_j, where positive survivor measure then forces the correlated branch. Divergent drift alone gives neither.

Theorem 5.9 (Translation)**.** *If all classes satisfy $a_i \equiv c \pmod{n_i}$ for one fixed integer $c \geq 0$, then $A = c + \{y : n_i \nmid y$ for all $i\}$ up to a finite set, where y runs over the positive integers, and $\delta_{\log}(A)$ exists and equals ν.*

Proof. For $x > c$ put $y = x - c \geq 1$. The class $(n_i, c \bmod n_i)$ kills x iff $n_i \mid y$ and $x \geq n_i$; and if $n_i \mid y$ then $y \geq n_i$, so $x \geq n_i$ holds automatically, the threshold mattering only at $x = c$, which is excluded. So apart from the finitely many $x \leq c$, $x \in A$ iff no n_i divides y. The set of such y has a logarithmic density by Davenport–Erdős, equal to the limit of the densities of its finite heads, which is ν; translation by c and a finite change preserve logarithmic density. □

Remark 5.10. The hypothesis $c \geq 0$ is necessary and the finiteness depends on it. The exceptions are the integers lying below their class's truncation point, that is the representatives of c modulo each n_i. For $c \geq 0$ and $n_i > c$ that representative is c itself, so at most $c + 1$ distinct integers arise however many classes there are. For $c < 0$ this fails badly: with $c = -1$ the representative modulo n is $n - 1 < n$, so *every* class contributes a fresh exception and the symmetric difference is infinite. Note also that an infinite aligned channel determines c uniquely, since two aligning integers would differ by a multiple of arbitrarily large moduli; so c cannot be normalised to be nonnegative. Theorem 5.11 is therefore proved below without appealing to this theorem.

Theorem 5.11 (Finitely many aligned channels)**.** *If the classes are a union of finitely many* aligned *channels then $\delta_{\log}(A)$ exists and equals ν.*

Proof. Let $C_1, \ldots, C_r$ be the channels, c_j the integer aligning C_j, and D_j the set of moduli occurring in C_j. We argue without any sign hypothesis on c_j, and in particular without invoking Theorem 5.9.

Fix N and let $D_j^{(N)}$ consist of the $\min\{N, |D_j|\}$ smallest members of D_j, so that $D_j^{(N)} = D_j$ once $N \geq |D_j|$; channels with $D_j^{(N)} = \varnothing$ contribute nothing and are omitted from the maximum defining $B(N)$ below. Let S_N be the survivor set of the finitely many classes of the channels whose moduli lie in the respective $D_j^{(N)}$, each channel being a set of classes and the classes assigned to it exhausted by modulus. Being a finite union of truncated residue classes, S_N is periodic apart from finitely many integers and has a density d_N equal to its logarithmic density.

We claim that for every x exceeding $B(N) := \max_j\bigl(\max D_j^{(N)} + |c_j|\bigr)$,

$$x \in S_N \setminus A \implies x - c_j \in M(D_j) \setminus M\bigl(D_j^{(N)}\bigr) \text{ for some } j,$$

where $M(D)$ is the set of positive multiples of members of D. Indeed such an x is killed by some class (n, a) of some C_j with $n \notin D_j^{(N)}$, so $n \mid x - c_j$, and $x > |c_j|$ makes $x - c_j$ positive, whence $x - c_j \in M(D_j)$. And x survives every head class, so for x larger than every head modulus we have $m \nmid x - c_j$ for all $m \in D_j^{(N)}$, that is $x - c_j \notin M(D_j^{(N)})$.

By Lemma 4.1, $\delta_{\log}\bigl(M(D_j) \setminus M(D_j^{(N)})\bigr) \to 0$ as $N \to \infty$, and logarithmic density is invariant under translation by a fixed integer of either sign, up to discarding finitely many initial terms

– which the bound $B(N)$ above has already excluded. Given $\varepsilon > 0$ choose N so that these r quantities total less than ε. Since $A \subseteq S_N$ and $S_N \setminus A$ is contained, apart from the at most $B(N)$ integers below the bound, in $\bigcup_j \big(c_j + (M(D_j) \setminus M(D_j^{(N)}))\big)$, the upper and lower logarithmic densities of A each lie within ε of d_N. As ε was arbitrary, $\delta_{\log}(A)$ exists; and since $d_N = \nu_N \downarrow \nu$ while the tail bound tends to 0, its value is ν. □

Remark 5.12. The hypothesis that the channels be *aligned* cannot be dropped from the proof: by Remark 2.4 an infinite channel need not admit an integer translate, and without one Theorem 5.9 does not apply. Whether the conclusion survives for merely compatible channels we do not know; such a channel has unbounded misaligned parts, beyond the reach of Theorem 7.2, and none of the later reductions is tailored to it.

Two structural facts produce compatible channels, which are aligned when finite and so feed Theorem 5.11 in that case. Pairwise coprime classes are automatically compatible, since $\gcd = 1$ makes the residue condition vacuous. And if a subfamily has $\gcd(n_i, n_j) = d$ for every pair, then partitioning it by residue modulo d yields at most d channels, one for each residue class that actually occurs, so divergent drift for the family forces divergent drift for one of them, carrying at least $1/d$ of the total.

6 Where Davenport–Erdős stops

The engine of the Davenport–Erdős argument [2] is the inequality

$$f(n)\log n \;\geq\; \sum_{d\mid n} f(d)\Lambda(n/d), \tag{1}$$

where f is the indicator of the killed set. Its proof uses one property and nothing else: the killed set is closed upward under divisibility, so $f(d) \leq f(n)$ whenever $d \mid n$, and the right side is at most $f(n)\sum_{d\mid n}\Lambda(n/d) = f(n)\log n$.

Proposition 6.1. *Shifted classes destroy that closure, and* (1) *fails. Consider the finite system with moduli* $2 \leq n \leq 39$ *and* $a_n = 1$*. Then* 31 *is killed while* 62 *is not, so the killed set is not closed upward under divisibility; and* (1) *fails at* $n = 42$*, where the left side is* 0 *while the divisors* 6*,* 14 *and* 21 *are all killed, giving a right side of* $\log 2 + \log 3 + \log 7 > 3.7$*.*

Remark 6.2. The restriction to a finite head is not a defect of the example but a feature of the setting. In the full system $(n, 1)$, $n \geq 2$, every $x \geq 3$ is killed by the class $n = x - 1$, so (1) holds trivially with equality. The Davenport–Erdős argument is applied to the finite heads U_k and passed to the limit, so it is exactly at the finite heads that (1) must hold, and exactly there that misalignment breaks it.

7 Bounded misalignment and cylinder decompositions

What survives is a splitting.

Definition 7.1. For a class (n, a) write $n = uw$ with $\gcd(u, w) = 1$ and $u \mid a$; call u the *aligned part* and the least such w the *misaligned part.* Then

$$x \equiv a \pmod{n} \iff u \mid x \ \text{ and } \ x \equiv a \pmod{w}.$$

Theorem 7.2. *If the misaligned parts all divide a fixed M, then $\delta_{\log}(A)$ exists and equals ν.*

Proof. Fix a residue c modulo M and write $C_c = \{x \equiv c\ (M)\}$. Since $w_i \mid M$, every $x \in C_c$ satisfies $x \equiv c \pmod{w_i}$, so class i can kill some element of C_c only when $a_i \equiv c \pmod{w_i}$: the cylinder *selects* which classes act on it. For the selected classes the remaining condition is $u_i \mid x$. Writing

$$D_c = \{u_i : \ a_i \equiv c \pmod{w_i}\},$$

the killed set inside C_c is therefore $M(D_c) \cap C_c$, up to the truncation.

Let $U = \bigcup_i K_i$ be the killed integers, U_r the union of the first r of them, and $B_r = \max_{i \le r} n_i$. If $x \in C_c$ lies in $U \setminus U_r$ and $x > B_r$, then some class kills it, so $x \in M(D_c)$; and no class of the head kills it, so, x exceeding every head modulus, $u_j \nmid x$ for every $u_j \in D_{c,r}$, where $D_{c,r}$ collects the selected u_j with $j \le r$. Hence

$$(U \setminus U_r) \cap C_c \ \subseteq\ \big(M(D_c) \setminus M(D_{c,r})\big) \cap C_c$$

apart from the finitely many $x \le B_r$.

Intersecting with C_c only shrinks a set, so the upper logarithmic density of the left-hand side is at most that of $M(D_c) \setminus M(D_{c,r})$, which tends to 0 as $r \to \infty$ by Lemma 4.1, the sets $D_{c,r}$ exhausting D_c. Summing the M cylinders, $\overline{\delta}_{\log}(U \setminus U_r) \to 0$.

Now given $\varepsilon > 0$ choose r with $\overline{\delta}_{\log}(U \setminus U_r) < \varepsilon$. The set U_r is a finite union of truncated progressions, hence periodic apart from finitely many integers, so it has a density $1 - \nu_r$. As $U_r \subseteq U \subseteq U_r \cup (U \setminus U_r)$, the upper and lower logarithmic densities of U each lie within ε of $1 - \nu_r$, so they agree. Hence $\delta_{\log}(U)$ exists and equals $\lim_r (1 - \nu_r) = 1 - \nu$, so $\delta_{\log}(A)$ exists and equals ν. ☐

Remark 7.3. Two more obvious routes to this theorem do not work, and we record them because each looks plausible. One cannot finish by applying Davenport–Erdős cylinderwise: a set of multiples intersected with a fixed residue class modulo M is not a set of multiples, and substituting $x = c + My$ turns divisibility into a congruence. Nor does the divisibility-monotone weight

$$\varphi(x) = \tfrac{1}{M}\#\{c \bmod M : \ u_i \mid x \text{ and } a_i \equiv c \pmod{w_i} \text{ for some } i\}$$

suffice, though Lemma 4.2 does give it a logarithmic mean: φ records the cylinders in which x *would* be killed, whereas x inhabits one cylinder, and the two quantities differ. For the classes $(6, 2)$, $(10, 4)$, $(14, 6)$, $(15, 5)$, $(21, 7)$ with $M = 105$, both limits are exact: the killed set is periodic modulo 210 with density $73/210 = 0.3476$, which is the limit of F, the harmonic mass of the killed set defined in Section 9; while φ is periodic modulo 70 and its logarithmic mean is $169/490 = 0.3449$. What makes the proof above work is that the cylinder *selects* the acting classes, which is exactly what $w_i \mid M$ provides.

The hypothesis can be relaxed, and the relaxation already reaches systems of unbounded misalignment.

Corollary 7.4. *Suppose the classes split as $S_1 \cup S_2$, where every misaligned part occurring in S_1 divides a fixed M, and $\sum_{i \in S_2} 1/n_i < \infty$. Then $\delta_{\log}(A)$ exists and equals ν. The misaligned parts of the system as a whole may be unbounded.*

Proof. Given $\varepsilon > 0$ choose N with $\sum_{i \in S_2,\ i > N} 1/n_i < \varepsilon$ and let T consist of S_1 together with the first N members of S_2. Every misaligned part occurring in T divides $M_1 = \operatorname{lcm}(M, w_{i_1}, \ldots, w_{i_N})$, which is finite, so Theorem 7.2 applies to T and $\delta_{\log}(\operatorname{surv} T) = \nu_T$. Now $A \subseteq \operatorname{surv} T$ and $\operatorname{surv} T \setminus A$ is contained in the union of the remaining members of S_2, whose upper logarithmic density is at most $\sum_{i \in S_2,\ i>N} 1/n_i < \varepsilon$ by the harmonic bound of Theorem 5.1. Hence the upper and lower logarithmic densities of A each lie within ε of ν_T, and likewise $|\nu_T - \nu| \le \varepsilon$; let $\varepsilon \to 0$. ☐

So unboundedness alone is not the obstruction: it must be unboundedness carrying divergent drift. For the system $(n, 1)$ the misaligned parts are $2, 3, \dots, n, \dots$, no finite M exists, and the cylinder decomposition is infinite.

A second extension reaches systems that Theorem 7.2 and Corollary 7.4 both miss. On the cylinder $x \equiv c \pmod{M}$, write $x = c + My$; by Lemma 3.3 the classes meeting the cylinder become a threshold system in y, the *induced system* at c, whose survivors are exactly the indices of the integers of the cylinder surviving the original.

Theorem 7.5 (Cylinder decomposition)**.** *Suppose there is a modulus M such that for every residue c modulo M the induced system at c either has profinite survivor measure* 0*, or has bounded misaligned parts. Then $\delta_{\log}(A)$ exists and equals ν.*

Proof. The sets $A \cap C_c$ for c modulo M are disjoint with union A, and there are finitely many, so it suffices that each has a logarithmic density. Fix c and let S be the survivor set of the induced system at c, so that $A \cap C_c = \{c + My :\ y \in S\}$. Then

$$\frac{1}{\log X} \sum_{t \in A \cap C_c,\ t \le X} \frac{1}{t} = \frac{1}{\log X} \sum_{y \in S,\ y \le (X-c)/M} \frac{1}{c + My} \ \longrightarrow\ \frac{1}{M}\, \delta_{\log}(S),$$

since $1/(c + My) = \bigl(1 + O(1/y)\bigr)/(My)$ and $\log\bigl((X-c)/M\bigr)/\log X \to 1$. By hypothesis $\delta_{\log}(S)$ exists. In the first case the induced system has profinite survivor measure 0, so $\overline{\delta}_{\log}(S) \le 0$ by Theorem 3.1 applied to that system, giving $\delta_{\log}(S) = 0$. In the second case Theorem 7.2 applies and gives $\delta_{\log}(S) = \nu_c$, the survivor measure of the induced system. Summing over c, and using $\sum_c \nu_c/M = \nu$, gives the result. ☐

The hypothesis asks more than the argument needs: it is enough that almost every cylinder be resolved.

Theorem 7.6 (Approximate cylinder decomposition)**.** *Suppose that for every $\varepsilon > 0$ there is a modulus M such that the residues c modulo M whose induced system is neither of profinite survivor measure* 0 *nor of bounded misaligned parts number fewer than εM. Then $\delta_{\log}(A)$ exists and equals ν.*

Proof. Fix ε and such an M, and split the residues into the resolved set G and the rest B, with $|B| < \varepsilon M$. Write C_G and C_B for the corresponding unions of cylinders, so that $A = (A \cap C_G) \sqcup (A \cap C_B)$. As in Theorem 7.5, $A \cap C_G$ is a finite disjoint union of sets each having a logarithmic density, so it has one; call it g. And $A \cap C_B \subseteq C_B$, a union of $|B|$ residue classes modulo M, of logarithmic density $|B|/M < \varepsilon$. Hence

$$g \ \le\ \underline{\delta}_{\log}(A) \ \le\ \overline{\delta}_{\log}(A) \ \le\ g + \varepsilon.$$

The value g depends on ε, but the gap bound does not; as ε was arbitrary the upper and lower logarithmic densities agree, and since g is the survivor measure of the resolved cylinders, which lies within ε of ν, the common value is ν. ☐

Remark 7.7. Theorem 7.5 is the case $\varepsilon = 0$. The relaxation is reachable where the strict form is not: a class of modulus n meets $M/\gcd(n, M)$ of the M residues, a share $1/\gcd(n, M)$ that equals $1/n$ once $n \mid M$ and is never smaller, so a fixed finite obstruction with moduli $n_1, \dots, n_k$ touches every cylinder for small M but, for M a multiple of their least common multiple, at most a share $\sum_j 1/n_j$ of them, which is below ε when the moduli are large though never 0. What the relaxed hypothesis still demands, and what remains unproved in general, is that the unresolved share can be driven to 0 at all.

Remark 7.8. Induced systems may repeat a modulus, whereas the systems of the introduction have strictly increasing ones. This costs nothing: the proof of Theorem 7.2 uses only the cylinder selection and Lemma 4.1, neither of which refers to distinctness.

Theorem 7.5 is not vacuous, and covers a system the earlier results do not. Take the doubled construction of Section 10: its misaligned parts are the small primes q and so are unbounded, and for any M the classes excluded from Corollary 7.4 are those with $q \nmid M$, whose drift $\sum_{q>Q} s/q$ diverges by Mertens – so neither Theorem 7.2 nor Corollary 7.4 applies. But at $M = 2$ the odd cylinder carries no classes at all, and the even cylinder's induced system has profinite survivor measure 0 – its blocks are independent and the product of their factors tends to 0 – so Theorem 7.5 gives its logarithmic density, which is 1/2.

One hope for verifying the hypothesis is that refining M might *starve* cylinders of drift until they die. It cannot: refinement is drift-invariant.

Proposition 7.9 (Drift invariance of refinement)**.** *For every modulus M, the average over the M cylinders of the induced drift equals the drift of the original system:*

$$\frac{1}{M} \sum_{c \bmod M} \sum_{i \text{ meeting } c} \frac{\gcd(n_i, M)}{n_i} = \sum_i \frac{1}{n_i}.$$

Proof. Write $g = \gcd(n_i, M)$. The class (n_i, a_i) meets the cylinder c exactly when $g \mid a_i - c$, which happens for M/g of the M residues, and on such a cylinder its induced modulus is n_i/g, contributing induced drift g/n_i. Its total contribution over all cylinders is therefore $(M/g) \cdot (g/n_i) = M/n_i$, independent of M. Summing over i and dividing by M gives the claim. □

Remark 7.10. So refinement conserves drift on average: it cannot make every cylinder drift-poor, though divergent drift may concentrate on a small proportion of cylinders, as the doubled construction of Section 10 shows with $M = 2$. Whatever verifies the hypothesis of Theorem 7.6 must come from the arithmetic of the residues, not from thinning the drift. This also explains why iterating the decomposition does not obviously terminate: the object being refined is invariant under the refinement.

A useful special case identifies the cylinders by a coprimality criterion rather than by inspection.

Corollary 7.11. *Suppose there is a modulus M such that, for every residue c modulo M, the induced system at c contains a pairwise-coprime subfamily of divergent drift. Then $\delta_{\log}(A)$ exists and equals* 0*.*

Proof. Fix c. By Theorem 5.4 applied to the induced system at c, its survivor set has density 0; the same product $\prod(1 - 1/n) \to 0$ is its profinite survivor measure, which is therefore 0 as well. So every cylinder satisfies the first alternative of Theorem 7.5, giving that $\delta_{\log}(A)$ exists, and each contributes 0. □

Remark 7.12. This settles the systems on which Proposition 9.4 is vacuous. Take the moduli to be the even numbers, each carrying two classes, one with an even and one with an odd residue, the residues otherwise in *general position* – here and below, residues drawn independently and uniformly at random, statements about which are measurements of one such draw unless a theorem is cited. No two moduli are coprime, so Theorem 5.4 does not apply to the system itself; but a class $(2m, a)$ with a even kills only even x, and writing $x = 2y$ its condition becomes $y \equiv b \pmod{m}$, so on the even cylinder the induced moduli run over *all* integers m and contain the primes. The classes with a odd do the same on the odd cylinder. Both cylinders then satisfy the hypothesis and the survivors vanish, as measured.

The hypothesis also correctly excludes the aligned families, which survive: for the even moduli at residue 0 the odd cylinder carries no classes at all, so it has no coprime subfamily of divergent drift, and the corollary does not apply – as it must not, since those survivors have density 1/2.

8 Reduction theorems

The results so far resolve a system cylinder by cylinder. The theorems of this section reduce a system to another one instead, and are the paper's principal positive results. A first mechanism exhausts cylinders rather than resolving them.

Theorem 8.1 (Exhaustion as reduction)**.** *Suppose the classes partition into groups $G_1, G_2, \dots$, where group G_j kills only integers in a residue class $C_j = c_j + d_j\mathbb{Z}$ (its* protector*), and suppose the induced system of G_j on C_j has profinite survivor measure* 0 *for every j. Call $\{(d_j, c_j)\}$ the* reduced system*, A' its survivor set and ν' its survivor measure. If $\delta_{\log}(A')$ exists and equals ν', then $\delta_{\log}(A)$ exists and equals $\nu' = \nu$.*

Proof. Write U' for the killed set of the reduced system, each reduced class (d_j, c_j) being given the threshold $\min(d_j, \min_{i \in G_j} t_i)$, at most d_j and, for all but the finitely many groups containing an exceptional class, at least d_j/M', so that the reduced system is a threshold system with the same constant. Since group j kills only inside C_j and only from its thresholds on, $U \subseteq U'$, so $\overline{\delta}_{\log}(U) \leq \overline{\delta}_{\log}(U')$. Since the profinite unions of the two systems agree up to a null set, $\nu' = \nu$, and by hypothesis $\delta_{\log}(U') = 1 - \nu' = \lim_k \delta\big(\bigcup_{j \leq k} C_j\big)$, the last equality because the finite unions have measures increasing to $1 - \nu'$. For the lower bound fix k: by Corollary 3.2 applied to the induced system on C_j, group j's survivors inside C_j have logarithmic density 0, so $U \supseteq \bigcup_{j \leq k} C_j$ up to a set of logarithmic density 0, giving $\underline{\delta}_{\log}(U) \geq \delta\big(\bigcup_{j \leq k} C_j\big)$ for every k. Letting $k \to \infty$ matches the upper bound. □

Remark 8.2. The aligned case – all $c_j \equiv c$ for one integer $c \geq 0$ – is the reduced system of Theorem 5.9; the sparse case $\sum 1/d_j < \infty$ is that of Theorem 5.1. The theorem may be iterated: the reduced system may itself be exhausted by its own grouping, reducing further. A *tower* of depth two is exhibited below, evading the single-level form.

Remark 8.3. This theorem is not a curiosity: it covers a system that *none* of the others reach, and which refutes the naive reduction discussed in Section 9. Take sparse primes p_j with $\sum 1/p_j < \infty$, and let G_j consist of the classes of modulus $p_j m$, for $m \geq 2$ coprime to p_j, with residue $\equiv 1 \pmod{p_j}$ and $\equiv 0 \pmod{m}$. Then G_j kills only inside $C_j = \{x \equiv 1\ (p_j)\}$, and within it kills every point divisible by some $m \geq 2$ coprime to p_j; the survivors in C_j are divisible by no prime other than p_j, a set of measure $\prod_{q \neq p_j}(1 - 1/q) = 0$, so G_j exhausts C_j. The moduli have greatest common divisor 1 and admit no finite prime cover, since p_j itself is a modulus of G_j; the residues are misaligned; each group's drift diverges; and yet $\nu = \prod_j (1-1/p_j) > 0$. The same holds, as a measurement, when the residues modulo m are instead taken in general position: with $p_j \in \{11, 197, 439, 691, 977, 1259\}$ the survivor density in the window $[X/2, X]$, $X = 4 \cdot 10^6$, is 0.8995, agreeing with the product to four decimals; in the three cylinders measured the survivor share is 0.0000, and in their complement 1.0000. So positive survivor measure against divergent misaligned drift does *not* require a common factor: many protectors, one per group, suffice.

Nor need the protecting cylinders be aligned. Take three sparse primes q_i, here $11, 197, 439$; for each, level-one classes $(q_i m, c)$ with $c \equiv 1 \pmod{q_i}$ and c in general position modulo m – a misaligned family of 172 cylinders – and let each be exhausted by its own group of level-zero classes. The level-zero system has gcd 1, drift 2.26, and 20,488 classes; its exhausted cylinders are not aligned, so a single application of Theorem 8.1 with the aligned reduced case does not reach it. Two applications do: the level-zero system reduces to the level-one family, which reduces in turn to $\{(q_i, 1)\}$, of convergent drift. The measured survivor density is 0.9024, agreeing with $\prod(1 - 1/q_i)$ to four decimals, the six level-one cylinders measured having survivor share 0.0000.

Exhaustion suggests a majorant that needs no partition into groups.

Theorem 8.4 (Saturation)**.** *Let the system be standard ($t_i = n_i$), and let $\mathcal{P}$ be the family of cylinders C with $\mu(\widehat{A} \cap C) = 0$ that are maximal under inclusion. Then every class lies inside a member of $\mathcal{P}$; the union of $\mathcal{P}$ is the union of the balls up to a null set, so the system $\mathcal{P}$ has survivor measure ν; $\mathcal{P}$ is* saturated, *in that no proper coarsening of a member has null survivor measure while every cylinder of null survivor measure lies inside a member; and $\mathcal{P}$ admits no proper null coarsening, so that the reduction applied to $\mathcal{P}$ returns $\mathcal{P}$. If the survivor set of $\mathcal{P}$ has logarithmic density equal to ν, then so does A.*

Proof. The cylinders containing a given cylinder $c + d\widehat{\mathbb{Z}}$ are its coarsenings $c + d'\widehat{\mathbb{Z}}$ with $d' \mid d$, finitely many; so among the coarsenings of a null cylinder that are themselves null there is a maximal one, and it is maximal under inclusion among all null cylinders, hence a member of $\mathcal{P}$. Applied to a class B_i, which is null since $\mu(\widehat{A} \cap B_i) = 0$, this puts every class inside a member; applied to an arbitrary null cylinder it gives the second half of saturation, and the first half is the maximality of the members. Every member of $\mathcal{P}$ meets $\widehat{A}$ in a null set and so lies in the union of the balls up to a null set, which gives the union and the measure. The profinite survivor set of $\mathcal{P}$ is $\widehat{A}$ up to a null set, and a null cylinder for one set is a null cylinder for the other, since $|\mu(C \cap \widehat{A}) - \mu(C \cap \widehat{A}')| \le \mu(\widehat{A} \triangle \widehat{A}')$; so the two sets have the same maximal null cylinders and the reduction is idempotent. For the density, a class inside $P \in \mathcal{P}$ kills only integers at least its modulus, which is a multiple of d_P, while P kills every integer at least d_P in its residue class; so the kills of $\mathcal{P}$ contain those of the original, $A \supseteq A_{\mathcal{P}}$, and $\underline{\delta}_{\log}(A) \ge \delta_{\log}(A_{\mathcal{P}}) = \nu$, with Theorem 3.1 for the other inequality. $\square$

Remark 8.5. Unlike Theorem 8.1, no partition of the classes into groups is involved – the members of $\mathcal{P}$ may meet without nesting – and the density statement is one-sided: the saturated system majorises the original, and a density for it passes down. The consequence is that in proving $\delta_{\log}(A) = \nu$ one may assume the system saturated: its classes are the maximal cylinders meeting the survivor set in a null set – "holes" up to measure zero, not necessarily disjoint from it – no proper coarsening is null, and any cylinder that dies is already inside a class. The converse direction, that a density for A forces one for $A_{\mathcal{P}}$, we have not proved and do not need. On finite systems modulo 1260 the reduction was computed exactly: a many-protector system of 1,062 classes collapses to its two protectors, a coprime family of four primes is fixed, and on six random systems every class lies in a member, the union is reproduced, the reduction is idempotent and its output saturated.

In the saturated form the classes are the maximal null cylinders of $\widehat{A}$, and Lemma 5.6, read backwards, says what positive survivor measure forces of them.

Corollary 8.6 (Anchoring)**.** *Let the system be saturated with $\nu > 0$, and for each P let $\mathcal{H}_P$ be the classes whose moduli are coprime to every prime at most P, and let ρ_P be the limsup of the second-moment ratios of $\mathcal{H}_P$ along the exhaustion by increasing modulus. Then either $\sum_{\mathcal{H}_P} \mu(C) < \infty$ for some P, in which case every class outside $\mathcal{H}_P$ lies in one of the finitely many cylinders $c + p\mathbb{Z}$ with $p \le P$; or $\sup_P \rho_P = \infty$.*

Proof. A class in $\mathcal{H}_P$ is determined by the coordinates at primes above P, and its points are all killed. If the sums diverged for every P and the ρ_P were bounded by one constant, Lemma 5.6 would put almost every point in infinitely many classes, so $\nu = 0$. The alternative in the first case is the definition of $\mathcal{H}_P$: a modulus not coprime to the primes up to P is divisible by one of them. □

Remark 8.7. So a saturated system with survivors either splits, at some level P, into escapees of summable measure – the summability hypothesis of Theorem 8.8, though not its covering hypothesis on finite unions – and classes anchored to a bounded family of prime cylinders, or its escapees are concentrated in a sense that strengthens without bound as P grows. Writing $N(x)$ for the number of classes containing x, the second alternative says $\int N^2 \gg (\int N)^2$ along the escapees: a set of small measure lies in an enormous number of holes. A counterexample must therefore either exhibit such concentration, or have its anchored classes – an infinite system confined to finitely many prime cylinders, possibly of divergent drift – fail the covering hypothesis of Theorem 8.8 there. We have not been able to convert either into one of the organisations covered above.

Exhaustion asks the groups to die. They need not, if the cylinders confining them are sparse.

Theorem 8.8 (Summable protectors)**.** *Suppose the classes partition into groups $G_1, G_2, \dots$, where group G_j kills only integers in a residue class $C_j = c_j + d_j\mathbb{Z}$, that $\sum_j 1/d_j < \infty$, and that for every J the survivor set A_J of $G_1 \cup \dots \cup G_J$ has logarithmic density equal to its profinite survivor measure ν_J. Then $\delta_{\log}(A)$ exists and equals ν.*

Proof. The profinite unions of the finite systems increase to that of the full system, so $\nu_J \downarrow \nu$, and by hypothesis $\delta_{\log}(A_J) = \nu_J$. Since $A \subseteq A_J$, $\overline{\delta}_{\log}(A) \le \nu_J$ for every J, hence $\le \nu$. Taking J past the finitely many groups containing an exceptional class, the groups beyond J kill only inside their cylinders, and only integers at least d_j/M' by the fourth threshold fact ($M' = 1$ for a standard system); so $A \supseteq A_J \setminus \bigcup_{j>J} \big(C_j \cap [d_j/M', \infty)\big)$. A residue class modulo d contributes at most $(\log X + \log M' + M')/d$ to the harmonic sum over $[d/M', X]$, so the union has upper logarithmic density at most $\sum_{j>J} 1/d_j$, and $\underline{\delta}_{\log}(A) \ge \nu_J - \sum_{j>J} 1/d_j$, which tends to ν. □

Remark 8.9. Theorem 5.1 is the case of singleton groups, each confined to its own ball, and Theorem 8.1 with summable protectors is the case of dying groups. The point is that the groups need not die: Proposition 9.12 below has divergent drift, positive survivor measure and no dying cylinder whatever, and this theorem covers it.

Both reductions are instances of one structure, and that structure can be pushed to its limit. A *tree of cylinders* is a family T of residue classes $c + d\mathbb{Z}$, $d \geq 1$, containing $\mathbb{Z}$, any two of which are nested or disjoint, each member having finitely many maximal proper sub-members, its *children*; a member's *gap* is what its children leave uncovered, and its *depth* is the number of its proper ancestors. A system is *coprime-layered* on T if

(T1) T is a tree of cylinders as just defined;

(T2) every class is $(d_C q, a)$ with $C = c + d_C\mathbb{Z}$ in T, q prime and $a \equiv c \pmod{d_C}$: the class is *attached* to C and lies inside it;

(T3) no class is contained in a class attached to a proper ancestor of its node;

(T4) no attached prime divides the modulus of any node.

Write $S_C = \sum 1/q$ over the classes attached to C. The nodes containing a point $x \in \widehat{\mathbb{Z}}$ form a chain, its *branch*, and $g(x) = \sum_C S_C$ over that chain is the *branch drift*; g_D denotes the sum over the nodes of depth at most D. Nothing here restricts the drift, the survivor measure, the alignment, or the depth.

Lemma 8.10 (Dying branches)**.** *In a coprime-layered system,* $\mu(\{g = \infty\} \cap \{\textit{survivors}\}) = 0$*.*

Proof. Fix a node C of depth D and a point $x \in C$ with index t, $x = c + d_C t$. A class attached to an ancestor C' of C, or to C itself, restricts on C to a single residue class of t modulo its prime, because by (T4) the prime is coprime to $d_C/d_{C'}$; by (T3) two classes on the chain with the same prime restrict to different residues, and so do two classes at the same node with the same prime, being distinct balls with the same residue modulo d_C. The index t is uniformly distributed modulo any product of these primes, so the points of C surviving the classes on the chain to C have measure $\mu(C)\prod_q(1-k_q/q) \leq \mu(C)e^{-g_D}$, with k_q the number of classes of prime q on the chain. The set $\{g_D > N\}$ is a union of nodes of depth at most D and of gaps, on each of which the same bound holds with the chain to that cell, so $\mu(\{g_D > N\} \cap \{\text{survivors}\}) \leq e^{-N}$. Since $\{g = \infty\} \subseteq \bigcup_D \{g_D > N\}$ for every N, and the sets on the right increase with D, the survivors in $\{g = \infty\}$ have measure at most e^{-N} for every N. □

Lemma 8.11 (Light and heavy)**.** *In a coprime-layered system, for every $N > 0$, $\underline{\delta}_{\log}(A) \geq \nu - \mu\{g > N\}$.*

Proof. Write g_C for the branch drift summed to C inclusive. Call a class *light* if $g_C \leq N$ at its node, and call a node a *first crossing* if $g_C > N$ while every proper ancestor has $g \leq N$. First crossings are pairwise disjoint, being non-nested members of T, and their union is exactly $\{g > N\}$. The light classes have total drift $\sum_C \mu(C)S_C$ over nodes with $g_C \leq N$, which is $\int \sum_{C \ni x,\ g_C \leq N} S_C \, d\mu(x) \leq N$, so by Theorem 5.1 their survivor set A' has logarithmic density $\nu' \geq \nu$. Every other class is attached to a node at or below a first crossing C_j, lies inside C_j, and kills only integers at least d_j/M', by the fourth threshold fact. Hence $A \supseteq A' \setminus \bigcup_j (C_j \cap [d_j/M', \infty))$, and as in Theorem 8.8 $\underline{\delta}_{\log}(A) \geq \nu' - \sum_j 1/d_j = \nu' - \mu\{g > N\} \geq \nu - \mu\{g > N\}$. □

Theorem 8.12 (Coprime-layered trees)**.** *For every coprime-layered system,* $\delta_{\log}(A)$ *exists and equals* ν*.*

Proof. The upper bound is Theorem 3.1. For the lower bound fix $\varepsilon > 0$ and $0 < \eta < \frac{1}{2}$. Let $\mathcal{F}_D$ be the finite partition of $\widehat{\mathbb{Z}}$ into the nodes of depth D and the residue classes composing the gaps of nodes of smaller depth, its *cells*; each cell is a single residue class, and the partitions refine as D grows, since a gap once formed is never subdivided. The dying set $E = \{g = \infty\}$ is determined by the branch, hence measurable for $\mathcal{F}_\infty = \sigma(\bigcup_D \mathcal{F}_D)$, so by Lévy's upward theorem [9] the conditional measure of E in the cell containing x tends to $1_E(x)$ for almost every x. Choose D so large that the cells in which the conditional measure of E lies in $(\eta, 1-\eta)$ – the *mixed* cells – have total measure at most ε; call the remaining cells *dying* or *alive* according as that measure is at least $1-\eta$ or at most η.

Let C be an alive cell, a residue class $c + d_C\mathbb{Z}$; either C is a node of depth D, or C is one of the residue classes composing the gap of a node C_0, in which case its modulus is a multiple of d_{C_0} built from node moduli. Write $x = c + d_C t$. The classes that meet C are those attached to the nodes containing C (its node and that node's ancestors) and, when C is a node, those attached to its descendants. By (T4) every attached prime q is coprime to d_C, so a class $(d_{C'}q, a)$ attached to a node $C' \supseteq C$ becomes in t the class of modulus q, attached to the root, and a class attached to a descendant $C'' \subset C$ becomes one of modulus $(d_{C''}/d_C)\,q$, attached to the node C'' of the subtree rooted at C. This is a coprime-layered system on that subtree (a single root when C is a gap class): (T1) and (T4) are inherited; (T2) holds with the induced node moduli; and (T3) holds because two root classes with the same prime q and the same residue modulo q would come from classes agreeing modulo q and, both containing C, modulo their node moduli, so that one would contain the other, which (T3) for the original system forbids. Its branch drifts are those of the original along the branches through C, its survivor set is the index set of $A \cap C$, and its survivor measure is $\nu_C = \mu(C \cap \{\text{survivors}\})/\mu(C)$. By Lemma 3.3 it is a threshold system, and Lemma 8.11 applies to it by the fourth threshold fact. Lemma 8.11 gives $\underline{\delta}_{\log}(A \cap C) \geq \mu(C)(\nu_C - \mu\{g > N \mid C\})$ for every N, and letting $N \to \infty$, $\underline{\delta}_{\log}(A \cap C) \geq \mu(C)\nu_C - \eta\mu(C)$. On a dying cell, Lemma 8.10 gives $\mu(C \cap \{\text{survivors}\}) \leq \eta\mu(C)$. Lower logarithmic density is superadditive over disjoint sets, so summing over the alive cells,

$$\underline{\delta}_{\log}(A) \;\geq\; \sum_{\text{alive}} \mu(C \cap \{\text{survivors}\}) - \eta \;\geq\; \nu - \eta - \varepsilon - \eta,$$

the survivors in dying cells having measure at most η and those in mixed cells at most ε. Let η and ε tend to 0. □

Remark 8.13. The theorem needs neither dying groups nor summable protectors. It covers, for instance, a tree whose deep cylinders retain measure bounded away from 0, whose nodes carry, in a proportion $1/k^2$ at depth k, finite drifts so large that the total drift diverges, and along almost all of whose branches the drift nonetheless converges, so that a positive measure of survivors lies at infinite depth: the case that Theorems 8.1 and 8.8 both miss. Its reach is nevertheless limited, and in a way that is easy to misjudge: being coprime-layered is a strong hypothesis on the moduli, not a normalisation.

Proposition 8.14 (Chain constraint)**.** *In a coprime-layered system, any two nodes carrying classes have non-coprime moduli, unless one of the moduli is* 1*.*

Proof. If $d, d' > 1$ are coprime then, by the Chinese remainder theorem, the classes $c + d\mathbb{Z}$ and $c' + d'\mathbb{Z}$ meet whatever c, c' are, so the two nodes are not disjoint and must be nested; but nesting forces $d \mid d'$ or $d' \mid d$, which with $\gcd(d, d') = 1$ makes one of them 1. □

So the classes of a coprime-layered system hang off a family of cylinders any two of which share a factor: the node moduli $2, 10, 100, 1700$ of the explicit tree described in Appendix A, not the moduli $7, 37, 73, \ldots$ of a family of protectors. Proposition 9.12 is therefore *not* coprime-layered – its natural nodes are the distinct primes p_j, pairwise coprime – and it is covered by Theorem 8.8 instead. Nor are the semiprimes, whose nodes would be the primes p; they are covered by Theorem 5.5. Checked by exact search over the legal node assignments on truncations – for the tree, its depth-two slice – that explicit tree and the multiples of 6 are coprime-layered, while those two systems are not.

Remark 8.15. Lemma 8.10 is the only place where primality of the attached moduli is used, which invites the thought that allowing a node to carry a layer of composite moduli would extend the theorem. It would not: the lemma's analogue for composite layers is false. Proposition 9.12 below is a system on the single cylinder $\widehat{\mathbb{Z}}$, with composite moduli, divergent drift and no proper reduction, whose survivor measure is positive; read as a layer it has $g = \infty$ everywhere and yet does not kill almost every point. So without primality the proof of Theorem 8.12 collapses at its first step, and an extension would have to establish, for each layer, the reverse inequality itself. The tree theorem is exactly as strong as the independence its coprimality provides.

The chain constraint says which nested families the tree theorem reaches. At the opposite extreme, protectors with pairwise coprime moduli, the difficulty disappears for a different reason: positive survivor measure itself produces the confinement.

Theorem 8.16 (Coprime protectors)**.** *Suppose the classes partition into groups G_j, where G_j consists of classes $(p_j q, a)$ with $a \equiv c_j \pmod{p_j}$ and q prime, the moduli p_j are pairwise coprime, the attached primes q are distinct across all classes, and none of them divides any p_i. Then $\delta_{\log}(A)$ exists and equals ν, whatever $\sum_j 1/p_j$ may be.*

Proof. Write $S_j = \sum 1/q$ over the primes attached in G_j and $\kappa_j = 1 - \prod(1 - 1/q)$, the measure of the kills of G_j inside $C_j = c_j + p_j\widehat{\mathbb{Z}}$ relative to $\mu(C_j)$. Whether x lies in C_j depends on x modulo p_j; whether $x \in C_j$ is killed by G_j depends on x modulo the primes attached in G_j. By hypothesis these coordinates are distinct across j, and the coordinates of $\widehat{\mathbb{Z}}$ are independent, so

$$\nu = \prod_j \Big(1 - \frac{\kappa_j}{p_j}\Big).$$

If $\sum_j \kappa_j/p_j = \infty$ then $\nu = 0$ and Corollary 3.2 finishes. Otherwise, since $\kappa_j \geq 1 - e^{-S_j} \geq (1 - e^{-1}) \min(S_j, 1)$, we get $\sum_j \min(S_j, 1)/p_j < \infty$: the groups with $S_j \geq 1$ lie on cylinders with $\sum 1/p_j < \infty$, and the remaining groups have total drift $\sum S_j/p_j < \infty$. A group with $S_j = \infty$ is a coprime layer of divergent drift, which dies on C_j by Theorem 5.4; apply Theorem 8.1 with those groups and every other class as a singleton group on its own ball, where its induced system is the single class of modulus 1 and dies trivially; this reduces each dying group to the class (p_j, c_j) and leaves every other class unchanged. The reduced system is covered by Theorem 8.8, with the groups: each reduced class and each group with $1 \leq S_j < \infty$, confined to its C_j, the moduli having $\sum 1/p_j < \infty$; and each remaining class as a singleton confined to its own ball, these measures summing to $\sum_{S_j<1} S_j/p_j < \infty$. A finite union of these groups has finite drift, so Theorem 5.1 gives its density as its survivor measure, the hypothesis of Theorem 8.8. □

Remark 8.17. This is the shape the checklist of Section 9 isolates – protectors too dense to be summable, kills too sparse to be quasi-independent – and the theorem says it cannot occur with coprime protectors: if the kills are sparse enough for survivors, the heavy protectors are already summable and the light classes already convergent. On an instance with 61 protectors, two of them carrying layers of drift 0.40 and the rest layers of drift $1/(40\log^2 p)$, all attached primes distinct, the product gives $\nu = 0.94479$ against a Monte Carlo value of 0.9413. Distinct primes make a single layer of drift 1 need primes beyond 10^9, so the instance checks the formula and the split, not divergence; for that, a layer of drift 1.5 on every odd prime has $\kappa \geq 1 - e^{-1.5}$, and the exact partial products $\prod_{p\leq N}(1-\kappa/p)$ fall through 0.2492 at $N = 10^3$, 0.1997 at $N = 10^4$, 0.1680 at $N = 10^5$, 0.1459 at $N = 10^6$, to 0 like $(\log N)^{-0.78}$. What the theorem does not reach is protectors that share factors without nesting – pairwise products $p_i p_j$, say – where membership is neither independent, as here, nor a chain, as in Theorem 8.12.

That last case is the remaining one, and it is settled by Lemma 5.6.

Theorem 8.18 (Overlapping protectors)**.** *Let $\mathcal{S}$ be a set of primes and, for every pair $p \neq p'$ in $\mathcal{S}$, let $G_{pp'}$ be a group of classes killing only inside a cylinder $C_{pp'}$ of modulus pp', whose kills there have relative measure $\kappa_{pp'}$. Assume that each group is a coprime layer of primes outside $\mathcal{S}$, distinct across groups, and that $\kappa_{pp'} \geq \kappa_0 > 0$. If $\sum_{p\in\mathcal{S}} 1/p$ diverges and every pair occurs, then $\nu = 0$ and $\delta_{\log}(A) = 0$. If it converges, then $\sum 1/(pp') < \infty$ and $\delta_{\log}(A) = \nu$ by exhaustion and confinement. Either way $\delta_{\log}(A)$ exists and equals ν.*

Proof. In the convergent case $\sum_{p<p'} 1/(pp') \leq \frac{1}{2}(\sum 1/p)^2$ is finite. A group whose layer has divergent drift dies on its cylinder by Theorem 5.4; apply Theorem 8.1 to those groups, with every other class a singleton, reducing each to the class of its cylinder. Then apply Theorem 8.8 to the reduced system with the groups confined to the cylinders $C_{pp'}$, of summable measure: a finite union of them has finite drift, so Theorem 5.1 covers it. Suppose then that $A := \sum_{p\in\mathcal{S}} 1/p = \infty$, and let $\widehat{K}_{pp'}$ be the kills of $G_{pp'}$, so $\mu(\widehat{K}_{pp'}) = \kappa_{pp'}/(pp')$. Delete from $\mathcal{S}$ the primes at most P and both endpoints of each of the finitely many groups whose attached primes include one at most P – finitely many because each prime is attached to at most one group. Finitely many primes are removed, so the reciprocal sum of the remaining primes still diverges, every pair of remaining primes still carries its group, and the remaining family is determined by the coordinates above P; enumerate it by the pairs among the first N remaining primes as N grows. For the second moment, two pairs sharing no prime give independent events, since the coordinates determining them – the two protector primes and the attached primes of the group, which lie outside $\mathcal{S}$ and are distinct across groups – are disjoint; and two sharing the prime p give $\mu(\widehat{K}_{pp'} \cap \widehat{K}_{pp''}) \leq \mu(C_{pp'} \cap C_{pp''}) = 1/(pp'p'')$. Writing A_N, B_N for $\sum 1/p$ and $\sum 1/p^2$ over the first N primes of $\mathcal{S}$ above P, and $M_N = (A_N^2 - B_N)/2$ for the total measure of the cylinders,

$$\sum_{e,f} \mu(C_e \cap C_f) = M_N + \sum_p \frac{1}{p}\Big[(A_N - 1/p)^2 - (B_N - 1/p^2)\Big] + \Big(M_N^2 - \Sigma'\Big),$$

where e, f run over ordered pairs of edges of the complete graph on those N primes, $\mu(C_e) = 1/pp'$ for $e = \{p, p'\}$, and the three terms are as follows. The diagonal $e = f$ contributes $\sum_e \mu(C_e) = M_N$. Two distinct edges $e = \{p, p'\}$, $f = \{p, p''\}$ sharing exactly the prime p meet in the cylinder of modulus $pp'p''$, and summing $1/pp'p''$ over ordered pairs $p' \neq p''$ of primes other than p gives $p^{-1}[(A_N - 1/p)^2 - (B_N - 1/p^2)]$; summing over p gives the middle term. Disjoint edges have independent cylinders, so their contribution is $\sum_{e\cap f=\varnothing} \mu(C_e)\mu(C_f)$, which is M_N^2 less the products over non-disjoint ordered pairs, $\Sigma' = \sum_e \mu(C_e)^2 + \sum_p p^{-2}[(A_N - 1/p)^2 - (B_N - 1/p^2)]$. Since $\widehat{K}_e \subseteq C_e$, the numerator for the kills is at most this sum; since $\mu(\widehat{K}_e) \geq \kappa_0\mu(C_e)$, the denominator is at least $\kappa_0^2 M_N^2$. The middle sum is at most A_N^3, $\Sigma' \le B_N^2 + A_N^2 B_N$, and $M_N \sim A_N^2/2$, so the displayed sum is $M_N^2(1+O(1/A_N))$, and the second-moment ratio of the kills is at most $\kappa_0^{-2}(1+O(1/A_N))$. So Lemma 5.6 applies with $\rho = 2\kappa_0^{-2}$, almost every point lies in infinitely many $\widehat{K}_e$, and $\nu = 0$; Corollary 3.2 finishes. $\square$

Remark 8.19. The ratio's approach to 1 is genuinely slow, since $A_N \asymp \log\log N$: over all primes up to $10^3, 10^4, 10^5, 10^6, 10^7$ the exact ratios are 3.72, 3.29, 3.03, 2.87, 2.74, while the normalised quantity $(R-1)A/4$ falls through 1.1553, 1.1345, 1.1217, 1.1129, 1.1065 toward the predicted 1. Over the same range the measure of the points lying in no protector at all falls through 0.4811, 0.3965, 0.3392, 0.2975, 0.2658, and at $N = 10^3$ agrees with a Monte Carlo value of 0.4767. The sparse instance p_i the least prime at least i^2 has, at 49 primes, total protector measure 0.1191 and protector-free measure 0.9070 against a Monte Carlo value of 0.9044: there the sums converge and confinement, not this theorem, is what applies.

9 The reduction

Recall $U = \bigcup_i K_i \subseteq \mathbb{N}$. For $k \ge 1$ let $U_k = \bigcup_{i \le k} K_i$ and

$$F(X) = \frac{1}{\log X} \sum_{x \in U,\ x \le X} \frac{1}{x}, \qquad F_k(X) = \frac{1}{\log X} \sum_{x \in U_k,\ x \le X} \frac{1}{x}.$$

Theorem 9.1. *If*

$$\sup_{X \ge 2} \big(F(X) - F_k(X)\big) \longrightarrow 0 \qquad (k \to \infty),$$

then $\delta_{\log}(A)$ exists and equals ν.

Proof. Each U_k is a finite union of truncated arithmetic progressions, hence periodic apart from finitely many integers, so $F_k(X) \to 1 - \nu_k$ as $X \to \infty$. Let $\varepsilon > 0$ and choose k with $\sup_{X \ge 2}\big(F(X) - F_k(X)\big) < \varepsilon$. Since $U_k \subseteq U$ we have, for every X,

$$F_k(X) \ \le\ F(X) \ \le\ F_k(X) + \varepsilon.$$

Letting $X \to \infty$ along the upper and lower limits gives

$$1 - \nu_k \ \le\ \liminf_X F(X) \ \le\ \limsup_X F(X) \ \le\ 1 - \nu_k + \varepsilon,$$

so $\limsup_X F - \liminf_X F \le \varepsilon$. As ε was arbitrary, $\lim_X F(X)$ exists; and since it lies within ε of $1 - \nu_k$ for every such k, while $\nu_k \downarrow \nu$, that limit is $1 - \nu$. Hence $\delta_{\log}(A) = 1 - \lim_X F(X) = \nu$. $\square$

It is worth recording an equivalent form in terms of window densities, since it moves the difficulty from an infinite tail to a set of scales. For $X \ge 2$ let

$$w(X) = \frac{\big|U \cap [X/2, X]\big|}{\big|\mathbb{N} \cap [X/2, X]\big|}$$

be the killed density in the dyadic window at X.

Lemma 9.2 (Window lower bound)**.** $\liminf_{X \to \infty} w(X) \ge 1 - \nu$.

Proof. Fix Y. The classes of modulus at most Y kill a periodic set of density $1-\nu_Y$ whose period L_Y is the lcm of those moduli. A window of length $X/2$ contains $\lfloor X/(2L_Y)\rfloor$ full periods and a remainder shorter than L_Y, so the density of that periodic set inside the window is $1-\nu_Y+O(L_Y/X)$. Since U contains it, $w(X)\ge 1-\nu_Y-O(L_Y/X)$. Let $X\to\infty$ with Y fixed to get $\liminf_X w\ge 1-\nu_Y$, then let $Y\to\infty$ and use $\nu_Y\downarrow\nu$. □

The reverse fails: classes of modulus close to X kill one element of the window apiece, so w can reach 1 while $1-\nu$ is far below (Section 10). Thus w is pinned from below and free above, and the whole question is how often it escapes.

For a set E of real scales $X\ge 1$ define its logarithmic density as the limit, when it exists, of $\frac{1}{\log Y}\int_{E\cap[1,Y]} dX/X$, with upper and lower versions as usual; $w(X)$ is defined for every real $X\ge 2$ by the window $[X/2,X]\cap\mathbb{N}$.

Theorem 9.3 (Excursion form)**.** *Writing $u=\log X$ and $w(v)$ for $w(e^v)$, one has $F(e^u)=\frac{1}{u}\int_{v=\log 2}^{v=u} w(v)\,dv+O(1/u)$, so $\delta_{\log}(A)$ exists and equals ν if and only if, for every $\delta>0$, the set*

$$E_\delta=\{X:\ w(X)>1-\nu+\delta\}$$

has logarithmic density 0 *in the sense just defined.*

Proof. With $S(X)=\sum_{x\in U,\ x\le X}1/x$ and $Y=e^u$, integrate w against dX/X over $[2,Y]$. A killed integer $x\le Y/2$ lies in the window $[X/2,X]$ exactly for $X\in[x,2x]$, and the window then holds $X/2+O(1)$ integers, so its weight is $\int_x^{2x}\frac{dX}{X}\cdot\frac{1}{X/2+O(1)}=\frac{1}{x}+O(x^{-2})$; a killed integer in $(Y/2,Y]$ receives at most $\int_x^Y\frac{2\,dX}{X^2}\le\frac{1}{x}$, and there are at most $Y/2$ of them, each weighted at most $2/Y$, so they contribute $O(1)$. Summing, $\int_{\log 2}^u w(v)\,dv=S(Y)+O(1)$, the errors $O(x^{-2})$ having a convergent sum. (The relation is not an exact identity: for $U=\mathbb{N}$ one has $w\equiv 1$ while $F=1+\gamma/\log X+o(1/\log X)$.) Hence $F(e^u)=\frac{1}{u}\int_{v=\log 2}^{v=u}w(v)\,dv+O(1/u)$, whose error vanishes, so F converges precisely when w has a logarithmic mean. By Lemma 9.2 the function $w-(1-\nu)$ is non-negative in the limit, so its average vanishes if and only if it is small off a set of vanishing logarithmic density, which is the stated condition. Finally $\delta_{\log}(A)=1-\lim F$. □

One part of the excess admits a quantitative bound. Write $D(Y)=\sum_{n_i\le Y}1/n_i$ and $h=\limsup_{Y\to\infty}D(Y)/\log Y$, the harmonic density of the modulus set; note that $h\le 1$ when the moduli are distinct.

Proposition 9.4 (Spike bound)**.** *Let $\sigma(X)$ denote the density in the window $[X/2,X]$ of the union of the kills of the classes of modulus in $(X/2,X]$, which bounds the contribution of those classes to $w(X)$ however the overlaps are apportioned. Then for every $\delta>0$ the set $\{X:\sigma(X)>\delta\}$ has logarithmic density at most $(2\log 2)\,h/\delta$. In particular it has logarithmic density* 0 *whenever $h=0$.*

Proof. A class of modulus $n\in(X/2,X]$ kills at most one element of $[X/2,X]$, since its elements are at least n and consecutive ones differ by $n>X/2$. Hence, with N the number of such classes and $X\ge 4$, $\sigma(X)\le N/(X/2-1)\le(2N/X)(1+4/X)$, the window holding at least $X/2-1$ integers. Each of these classes contributes at least $1/X$ to the window drift $d(X)=\sum_{X/2<n_i\le X}1/n_i$, giving $N/X\le d(X)$ and therefore $\sigma(X)\le 2(1+4/X)\,d(X)$.

Since $n_i\in(X/2,X]$ exactly when $X\in[n_i,2n_i)$,

$$\int_1^Y d(X)\,\frac{dX}{X}=\sum_{n_i<Y}\frac{1}{n_i}\int_{n_i}^{\min(2n_i,Y)}\frac{dX}{X}\le(\log 2)\,D(Y),$$

so for $\delta' < \delta$ Markov's inequality for the measure dX/X on $[1, Y]$ bounds the dX/X-measure of $\{X \le Y : 2d(X) > \delta'\}$ by $(2\log 2)D(Y)/\delta'$, whose ratio to $\log Y$ has limsup $(2\log 2)h/\delta'$. Since $\sigma \le 2(1 + 4/X)d$, the set $\{\sigma > \delta\}$ lies in $\{2d > \delta'\}$ beyond a bounded X; let $\delta' \to \delta$. $\square$

Remark 9.5. The scope is exactly the spike term. Classes of *intermediate* modulus, say $T < n_i \le X/2$, are not controlled this way: bounding their contribution by $\sum 1/n_i$ over that range gives about $h \log X$, and restricting to the survivors of the small-modulus classes improves this only to $\nu_T\, h \log X$. Both are useless. What Proposition 9.4 does show is that the mechanism of Section 10 – fresh classes of modulus comparable to X – can produce excursions on a set of positive logarithmic density only when the moduli have positive harmonic density.

The natural alternative also fails, and for a classical reason worth recording. One might hope that the window $[X/2, X]$ samples the union of the intermediate classes at its global density. It does not: that union is periodic with period the lcm of the moduli involved, of size roughly $e^{X/2}$, vastly longer than the window. The discrepancy is not hypothetical. Take the intermediate classes to be the primes $p \le X/2$ with residue 0. Their survivors inside $[X/2, X]$ are exactly the primes there, of density $\sim 1/\log X$, whereas the global survivor density of the same finite system is $\prod_{p \le X/2}(1 - 1/p) \sim e^{-\gamma}/\log X$ by Mertens. The window therefore over-represents survivors by a factor tending to $e^{\gamma} = 1.781\ldots$; numerically the ratio runs 1.7213, 1.7278, 1.7342, 1.7388 for $X = 2 \cdot 10^5$, 10^6, $4 \cdot 10^6$, $1.6 \cdot 10^7$. This is the Buchstab phenomenon, and it shows that no argument based on the window inheriting global densities can control the intermediate range. We regard this as the precise reason that range resists, rather than as evidence about the answer.

One might hope that the discrepancy always runs in the harmless direction. In the example above the window retains *more* survivors than the global density predicts, so the intermediate classes *under*-kill, and under-killing cannot produce an excursion. If that were general, Proposition 9.4 would control the whole excess. It is not general. Taking the classes to be the primes in $[X^a, X^b]$ at residue 0, for which coprimality makes the global survivor density the exact product $\prod(1-1/p)$, the ratio of the window survivor density to the global one is measured at $X = 8 \cdot 10^6$ as

$$0.9949 \quad (a, b) = (0.15, 0.35), \qquad 0.9107 \quad (a, b) = (0.05, 0.50),$$

both below 1, against 1.0174 and 1.0549 for $(0.20, 0.45)$ and $(0.25, 0.50)$. So the ratio crosses 1 in both directions, as the oscillation of Buchstab's function suggests, and intermediate classes can genuinely over-kill. The spike bound alone therefore does not control the excess.

The excess admits a clean expression in terms of the quantity sieve theory studies. Let $s(X)$ be the survivor density in the window and ν_X the profinite survivor measure of the classes of modulus at most X, and set $r(X) = s(X)/\nu_X$ – the window-to-global survivor ratio, the Buchstab ratio of Remark 9.5.

Proposition 9.6 (Reduction to the Buchstab ratio)**.** *Assume $\nu > 0$. For every X,*

$$w(X) - (1 - \nu) \;\le\; \nu\,\bigl(1 - r(X)\bigr)^{+},$$

so there is no excursion at all when $r(X) \ge 1$. Consequently $E_\delta \subseteq \{X : r(X) < 1 - \delta/\nu\}$, and $\delta_{\log}(A)$ exists and equals ν as soon as $(1-r)^{+} \to 0$ in logarithmic mean; since $\limsup_X r(X) \le 1$ *(shown below), this is the same as $r \to 1$ in logarithmic mean.*

Proof. The excess is $w(X) - (1-\nu) = \nu - s(X) = \nu - r(X)\nu_X$. If $r(X) \geq 1$ then, as $\nu_X \geq \nu$, this is at most $\nu - \nu_X \leq 0$. If $r(X) < 1$ then $r\nu_X \geq r\nu$ and the excess is at most $\nu(1-r)$. The remaining statements follow from Theorem 9.3. $\square$

Remark 9.7. This also explains why the spike term yielded to Proposition 9.4 while the intermediate range did not. A class of modulus comparable to X appears in $O(1)$ windows, so the window drifts sum to $D(Y)$ and form a budget that Markov's inequality can exploit. A class of intermediate modulus appears in *every* larger window, so no such budget exists and no Markov bound is available. Numerically, among nine dyadic windows up to $2 \cdot 10^6$, a modulus of size 10^6 occurs in two of them and a modulus of size 10^3 in all nine.

Two observations narrow what must be shown. First, an excursion requires $\nu > 0$: the excess is $\nu - s(X)$ and $s(X) \geq 0$, so a system whose survivors vanish can never produce one. This removes every dying system, and in particular disposes of the alarming ratio of Remark 9.5: the prime sieve has $\nu = 0$, so its e^γ discrepancy threatens a bound, never the conjecture. (At finite X the quantity ν_X is still $e^{-\gamma}/\log X$ and far from its limit, which is easy to confuse with ν itself.)

Second, the deviation is one-sided. For each fixed N the survivors of the whole system lie inside those of the first N classes, whose window density tends to ν_N; hence $\limsup_X s(X) \leq \nu_N$ for every N, and letting $N \to \infty$, $\limsup_X s(X) \leq \nu$. Since $\nu_X \to \nu > 0$, it follows that $\limsup_X r(X) \leq 1$, which is what makes the signed and the one-sided logarithmic means in Proposition 9.6 equivalent. Combining, the entire remaining content of the problem is that

$$r(X) \to 1 \ \text{ in logarithmic mean, for systems with } \nu > 0.$$

Numerically, on a system with ν known exactly the ratio sits at 1 to three decimals across every window, the largest value being 1.000136; by contrast the *sub*-systems of Remark 9.5 reach 1.3297, so the one-sidedness is a property of full systems and not of sieving in general.

The surviving hypothesis $\nu > 0$ is itself structural.

Proposition 9.8 (Structure of the open case)**.** *Suppose $\nu > 0$ and let $\varepsilon > 0$. Then there is a modulus M such that the residues c whose cylinders are at least $(1-\varepsilon)$-full of survivors carry all but ε of the survivor measure, and for each such c the induced system at c contains* no *pairwise-coprime subfamily of divergent drift.*

Proof. The survivor set $\widehat{\mathbb{Z}} \setminus \widehat{U}$ is closed of measure $\nu > 0$, so by the Lebesgue density theorem in $\widehat{\mathbb{Z}}$ almost every one of its points has cylinders about it of survivor proportion tending to 1; choosing M large enough along the divisibility order gives the first assertion. For the second, fix such a c. If the induced system at c had a pairwise-coprime subfamily of divergent drift, then by Theorem 5.4 its survivor measure would be 0, contradicting that the cylinder is $(1-\varepsilon)$-full. $\square$

Remark 9.9. This says exactly why Theorem 5.4 cannot reach the open case: the hypothesis $\nu > 0$ forbids, on the very cylinders carrying the survivors, the coprime divergence that theorem needs. The separation is sharp in practice. For the system with moduli p^2, where $\nu = \prod_p (1 - p^{-2}) = 6/\pi^2 = 0.6079\ldots$, the induced coprime drift on every cylinder modulo 6 is finite, being at most $\sum_p 6/p^2$ plus the contributions of 2 and 3; for the system of all moduli, where $\nu = 0$ whatever the residues (Theorem 5.4, the primes being among the moduli), it diverges on every cylinder.

This suggests a question. Since $\nu > 0$ forbids a pairwise-coprime subfamily of divergent drift (Theorem 5.4), the moduli must share primes heavily; and if every modulus were divisible by one of finitely many primes, a coprime subfamily could have at most that many members. The two simplest systems with $\nu > 0$ and divergent drift – the even moduli at residue 0 and the doubled construction – are indeed covered by the single prime 2.

The naive reduction – that divergent drift with $\nu > 0$ should force the moduli to be covered, up to a subfamily of convergent drift, by finitely many primes, or the residues to be aligned – fails: the many-protector system of the remark following Theorem 8.1 has $\nu > 0$, divergent drift, no finite prime cover and misaligned residues, and its density exists by cylinder exhaustion. The question that would close the problem is therefore the following.

Question (refined). *Can a system with divergent drift and $\nu > 0$ pass the checklist below in full – that is, satisfy the hypotheses of none of the theorems it names, nor reduce by finitely many applications of Theorem 8.1 to one that does?*

One kind of counterexample would be an *infinite tower*: a system each of whose reductions has divergent drift, positive survivor measure and misaligned residues, never terminating in a covered one; another would admit no proper reduction at all and fail every other hypothesis on the list. Each reduction passes to the moduli of the protecting cylinders, which are divisors of the previous level's moduli and must be sparse enough that their union leaves survivors; whether infinitely many such levels can each carry divergent drift is the question.

Three constraints on any such tower are immediate.

Proposition 9.10 (Tower constraints)**.** *In a tower of proper reductions – each level coarsening infinitely many classes – the following hold.*

- *(i) A class of level-zero modulus n_0 can be coarsened at most $\log_2 n_0$ times, since each proper coarsening passes to a proper divisor and at least halves the modulus; so among the classes coarsened at level K, only those of level-zero modulus $n_0 \geq 2^K$ occur. (A class left uncoarsened – a singleton group on its own ball – persists unchanged, so this constrains the coarsened classes, not the whole system.)*
- *(ii) No level contains a modulus equal to 1, for that cylinder is all of $\widehat{\mathbb{Z}}$ and its exhaustion forces $\nu = 0$.*
- *(iii) No non-top level consists of prime moduli, since a prime has no proper divisor exceeding 1; the classes that persist through K levels have moduli with at least K prime factors counted with multiplicity.*

Proof. Each is a direct reading of the definition of a proper reduction: the reduced modulus is a proper divisor of every modulus in its group. □

So an infinite tower needs, at every level, infinitely many classes still being coarsened, hence level-zero classes with unboundedly long divisor chains, each level of divergent drift and positive survivor measure, and no level reducing to primes. Attempting to build one exhibits a dichotomy. Take moduli $2^a q$ over sparse primes q and unbounded a, with residues aligned within each q (the only arrangement leaving survivors); level K has moduli $2^{a-K}q$, drift $\sim \sum 1/q$ independent of K, and $\nu = \prod(1 - 1/2q)$. If $\sum 1/q < \infty$ then $\nu > 0$ but every level has convergent drift, so the tower terminates in Theorem 5.1. If $\sum 1/q = \infty$ then every level has divergent drift but $\nu = 0$, and Corollary 3.2 settles it. Over twelve sparse primes, $\nu = 0.9535$ with drift 0.093 at every level; over four hundred dense primes, drift 2.33 with ν at 0.29, and $\nu \to 0$ as primes are added, by Theorem 5.4. The two conditions were not obtainable together.

We offer the reason as a heuristic, not a proof: coarsening divides out the structure the classes share, so the moduli at the top of a tower tend toward pairwise coprimality, and there Theorem 5.4 forces exactly this dichotomy – divergent drift kills the survivors, while positive survivor measure forces the drift to converge. If that heuristic could be made rigorous, towers would always be finite and, by the refined Question, the problem would close.

The heuristic predicts that coarsening raises coprimality. Under the canonical coarsening $n \mapsto n/p(n)$, dividing out the smallest prime, two measures over five levels were computed: the fraction of modulus pairs with gcd 1, and the drift share of the largest pairwise-coprime subfamily. For the even moduli the pair fraction rose from 0 to 0.63; for all integers from 0.61 to 0.64; for products $2^a 3^b q$ from 0 to 0.74. A family built to resist – the multiples of 6, sharing two primes – stayed at 0 for one step, since removing 2 leaves 3 in common, then rose to 0.59: a system sharing k primes needs k steps, which the heuristic accommodates. The coprime-subfamily share rose in every family, from between 0.12 and 0.35 to between 0.45 and 0.75. This is empirical support for the direction of the heuristic and nothing more; a proof must control the rate, since a tower could in principle coarsen forever without ever becoming coprime enough.

The dichotomy just described is false at a single level: the semiprimes have divergent reciprocal sum, no common factor, and every pairwise coprime subfamily convergent, since such a subfamily is a matching on the primes and the best matching $6, 35, 143, \ldots$ converges. Whatever closes the problem must come from coarsening, not from the shape of one level. The following system has divergent drift, misaligned residues and no proper exhaustion step, and Theorem 5.5 settles it.

Proposition 9.11 (Irreducible semiprimes)**.** *For each prime p let M_p consist of the primes above p taken in increasing order until their reciprocal sum S_p reaches $1/2$, and take the classes (pq, a_{pq}), $q \in M_p$, with arbitrary residues. Then:*

- *(i) every cylinder (d, c) with $d \geq 2$ containing two or more classes has $d = p$ prime, its induced moduli are distinct primes, its induced drift is bounded, and its induced survivor measure is positive; so no group of two or more classes dies in the sense of Theorem 8.1, and the system admits no proper exhaustion step, only the trivial ones in which a class is its own group;*
- *(ii) the drift diverges;*
- *(iii) $\nu = 0$ for every choice of residues.*

Proof. (i) A cylinder whose modulus has two prime factors contains only the class with that modulus. For $d = p$ the classes are $(pq, \cdot)$ with $q \in M_p$ and $(p'p, \cdot)$ with $p \in M_{p'}$; the induced moduli q and p' are distinct primes, so the induced survivor measure is $\prod(1 - 1/\ell)$ over them, positive when $\sum 1/\ell$ converges. That sum is at most $S_p + \sum_{p' : p \in M_{p'}} 1/p'$; the first term is below $1/2 + 1/p$. For the second, $p \in M_{p'}$ holds iff $\sum_{p' < q < p} 1/q < 1/2$, a condition that only becomes easier as p' increases, so the admissible p' are the primes in an interval $[p'_0, p)$, and their reciprocal sum is $1/p'_0 + \sum_{p'_0 < q < p} 1/q < 1/p'_0 + 1/2 \leq 1$. (ii) The drift is $\sum_p S_p/p \geq \frac{1}{2} \sum_p 1/p$. (iii) The smallest prime of pq is p. Writing $M_{p,c}$ for the $q \in M_p$ with $a_{pq} \equiv c \pmod{p}$ and $F_{p,c}$ for the event that some $q \in M_{p,c}$ has $x \equiv a_{pq} \pmod{q}$,

$$\mu(\widehat{K}_p) = \sum_c \frac{1}{p}\Big(1 - \prod_{q \in M_{p,c}} (1 - 1/q)\Big) \;\geq\; \frac{1}{p} \sum_c \big(1 - e^{-S_{p,c}}\big) \;\geq\; \frac{1 - e^{-1/2}}{p},$$

the last step because $(1-e^{-a})+(1-e^{-b}) \geq 1-e^{-a-b}$. So $\sum_p \mu(\widehat{K}_p)=\infty$. For $p \neq p'$, $\mu(\widehat{K}_p \cap \widehat{K}_{p'}) = \sum_{c,c'} \mu(F_{p,c} \cap F_{p',c'})/pp'$. Say $p < p'$, and condition on $x \equiv c \pmod p$ and $x \equiv c' \pmod{p'}$, an event of measure $1/pp'$. Given it, $F_{p,c} \cap F_{p',c'}$ is contained in the union of three events. E_0: $p' \in M_{p,c}$ and $a_{pp'} \equiv c' \pmod{p'}$, so that the witness $q = p'$ for the p-group is supplied by the conditioning itself; this can happen for only one pair (c, c'), namely $c \equiv a_{pp'} \pmod p$, $c' \equiv a_{pp'} \pmod{p'}$, and contributes at most $1/pp'$ in all. E_1: a common witness $q \in M_{p,c} \cap M_{p',c'}$, $q \neq p'$, with $a_{pq} \equiv a_{p'q} \pmod q$; given the conditioning the coordinate at q is still uniform, so E_1 has measure at most $\sum 1/q$ over such q, and each $q \in M_p \cap M_{p'}$ belongs to exactly one pair (c, c'), so the E_1 terms total at most $\min(S_p, S_{p'})/pp'$. E_2: two distinct witnesses $q \in M_{p,c} \setminus \{p'\}$ and $q'' \in M_{p',c'}$ with $q \neq q''$; both lie outside $\{p, p'\}$ (as $M_{p'}$ lies above p'), so given the conditioning their coordinates are independent and uniform, and the union bound over pairs gives measure at most $S_{p,c}S_{p',c'}$, whose sum over (c, c') is $S_pS_{p'}$. Altogether $\mu(\widehat{K}_p \cap \widehat{K}_{p'}) \leq (S_pS_{p'} + \min(S_p, S_{p'}) + 1)/pp' \leq (7/4 + o(1))/pp'$, so the ratio of Theorem 5.5 is at most $(7/4)/(1-e^{-1/2})^2 + o(1) < 11.4$. □

With $p \leq 150$ the system has 7,174 classes on 582 primes up to 4,253; computed exactly, the ratio is 1.889 at $N = 149$ and falling from 2.159, the least value of $p\,\mu(\widehat{K}_p)$ is 0.4815 against the bound 0.3935, the largest induced drift on any cylinder is 0.886, and the least induced survivor measure 0.316. Restricting to four sparse values of p makes $\sum \mu(\widehat{K}_p) = 0.061$, the theorem falls silent as it must, and the survivors have measure at least 0.939; a Monte Carlo draw of 3,000 points of $\widehat{\mathbb{Z}}$ returned 0.941. So irreducibility does not yield a counterexample: a top of a tower with $\sum_p \mu(\widehat{K}_p) = \infty$ must carry smallest-prime kills whose second-moment ratio, along the exhaustion by smallest prime, is unbounded. Such correlation does *not* force a dying cylinder.

Proposition 9.12 (Irreducible with survivors). *Let $p_1 < p_2 < \cdots$ be primes with $\sum_j 1/p_j < 1$, let $S_j \to \infty$ with $\sum_j S_j/p_j = \infty$ – for instance p_j the $\lceil 4j^{3/2} \rceil$th prime and $S_j = \sqrt{j}$: then $\sum_j 1/p_j < 0.24$, using $p_k > k \log k$ beyond the first six terms, while $p_k \sim k \log k$ gives $\sqrt{j}/p_j \sim 1/(6j \log j)$, whose sum diverges – and let group j consist of the classes (p_jq, a) with $a \equiv 1 \pmod{p_j}$ and a arbitrary modulo q, over primes q outside $\{p_i\}$ taken in order until $\sum 1/q$ reaches S_j. Then the drift diverges, $\nu \geq 1 - \sum_j 1/p_j > 0$, no cylinder holding two or more classes has induced survivor measure 0 (for the cylinder $\widehat{\mathbb{Z}}$ itself this is $\nu > 0$), so Theorem 8.1 admits no proper reduction of the system; and $\delta_{\log}(A) = \nu$ by Theorem 8.8.*

Proof. The drift is $\sum_j S_j/p_j$. All kills lie in $\bigcup_j C_j$ with $C_j = \{x \equiv 1\ (p_j)\}$, giving the bound on ν. A cylinder of modulus at least 2 holding two classes has prime modulus: for $(p_j, 1)$ the induced classes have the distinct prime moduli q of group j, with $\sum 1/q < S_j + 1$; for (q, c) they have the distinct prime moduli p_j of the groups using q with $a \equiv c \pmod q$, with sum below 1; in both cases the induced survivor measure is $\prod(1 - 1/\ell) > 0$. Cylinders of composite modulus hold at most one class. Finally $G_1 \cup \cdots \cup G_J$ has drift $\sum_{j \leq J} S_j/p_j < \infty$, so Theorem 5.1 covers it, and $\sum_j 1/p_j < \infty$. □

With six groups the system has 6,500 classes; every cylinder holding two or more classes has prime modulus and distinct prime induced moduli, the least induced survivor measure is 0.0637, the largest induced drift $2.449 = \sqrt{6}$, $\sum_{j \leq 6} 1/p_j = 0.1999$, and 6,490 classes are misaligned, with misaligned parts growing without bound. Replacing the protectors by the first primes, so that $\sum 1/p_j$ diverges, sends the survivors down through $0.487, 0.432, 0.391, 0.359$ as $5, 10, 20, 40$ groups are added, as the theorem's failure predicts.

So the tower picture is incomplete in the other direction too: a top that no exhaustion touches can be covered by confinement, and the infinite protector tree that evades both – deep cylinders of measure bounded away from 0, drift divergent in total yet convergent along almost every branch – is covered by Theorem 8.12. What a counterexample must now evade is the coprime-layered structure itself. By Proposition 8.14 that structure is a genuine restriction: the nodes must share factors pairwise, so the systems it reaches are those whose classes hang off a single divisor chain of cylinders, one prime at a time. Two natural hopes are therefore closed off. Not every system admits a coprime-layered refinement – the protector systems and the semiprimes do not, though other theorems cover them – and relaxing coprimality of the layers loses the dying-branch lemma outright, by Remark 8.15. The shape one might then fear – moduli spread across pairwise coprime cylinders with kills neither summable nor correlated – is ruled out by Theorem 8.16, where positive survivor measure forces the confinement. Nested protectors are the tree theorem's; pairwise coprime protectors are this one's. Protectors that share factors without nesting are Theorem 8.18's, and there the dichotomy is sharp: the protecting primes either have divergent reciprocal sum, and then almost every integer is caught, or convergent, and then the protectors are summable. All three regimes – nested, coprime, overlapping – are therefore closed, each under the hypotheses of its theorem: finitely many children and coprime layers for the trees, distinct attached primes for the coprime protectors, every pair of the prime set with kills bounded below for the overlapping ones. What is not closed is the passage from these regimes to an arbitrary system. Each theorem assumes a given organisation of the classes into groups with a stated relation between their moduli, and no result here produces such an organisation from nothing. That is too generous to the organisation theorems, whose hypotheses are strong: Theorem 8.16 wants its attached primes distinct across all classes, so even Proposition 9.12, which reuses them, is covered by confinement rather than by it; and the semiprimes are covered by Theorem 5.5 without any organisation at all. The final form is therefore the refined Question itself, read as a checklist. A counterexample to the existence of the logarithmic density would have to satisfy all of the following at once.

1. positive survivor measure (Corollary 3.2);
2. divergent drift (Theorem 5.1);
3. no pairwise coprime subfamily of divergent drift (Theorem 5.4);
4. smallest-prime kills whose measures are summable, or whose second-moment ratio along the exhaustion by smallest prime is unbounded (Theorem 5.5);
5. for every modulus M, some cylinder modulo M whose induced system has no pairwise coprime subfamily of divergent drift (Corollary 7.11);
6. no representation as finitely many aligned channels (Theorems 5.9, 5.11);
7. unbounded misaligned parts, not confined to classes of convergent drift (Theorem 7.2, Corollary 7.4);
8. for some $\varepsilon_0 > 0$ and every modulus M, at least a proportion ε_0 of the cylinders modulo M neither die nor have bounded misaligned parts (Theorems 7.5, 7.6);
9. no grouping into dying cylinders whose reduced system has density equal to its survivor measure (Theorem 8.1); and the same for its saturated form, since by Theorem 8.4 a saturated form with $\delta_{\log}(A_{\mathcal{P}}) = \nu$ would force $\delta_{\log}(A) = \nu$;

10. no grouping into cylinders of summable measure with covered finite unions (Theorem 8.8);
11. no coprime-layered structure (Theorem 8.12);
12. no pairwise coprime protectors with distinct attached primes (Theorem 8.16);
13. no protectors running over every pair of a set of primes, carrying coprime layers of distinct attached primes with kills bounded below (Theorem 8.18).

Every system known to us fails this checklist at some line. Whether it can be passed in full is the problem.

Theorem 8.4 gives a sufficient form without congruences: a positive answer to the following question for every closed set would settle the problem, though a negative one need not refute it. Let $\widehat{A} \subseteq \widehat{\mathbb{Z}}$ be closed, of positive Haar measure, and let $\mathcal{H}$ be the cylinders meeting $\widehat{A}$ in a Haar-null set that are maximal under inclusion, of divergent total measure. For an integer x, say x is *caught* if $x \geq d$ for some $c + d\mathbb{Z} \in \mathcal{H}$ containing it. Is the logarithmic density of the integers not caught equal to $\mu(\widehat{A})$? Corollary 8.6 says the holes escaping any finite set of primes are either summable or concentrated; the theorems above settle the cases in which the holes are nested, pairwise coprime, or pairwise products, each under the hypotheses of Theorems 8.12, 8.16 and 8.18. What remains is a closed set whose holes are concentrated in none of those ways, and whether such a set exists is the form in which we leave the problem.

The role of the common factor can be isolated exactly. Take the odd moduli $3, 5, 7, \ldots$ with residues in general position: unbounded prime support, greatest common divisor 1, drift $\sim \frac{1}{2} \log X$, and no alignment. Its survivors fall, not quite monotonically, to 0.0033 at $X = 4 \cdot 10^6$. Now double every modulus and force the residues even, changing nothing except that every kill is confined to the even numbers: the survivors sit at 0.5016, the odd numbers being structurally unreachable. The two systems differ only by the protector, and only the protected one retains survivors. In its first form, before refinement, the question asked whether positive survivor measure against divergent misaligned drift always requires such a protector, hence a common factor; the many-protector system answers that in the negative, and the refined Question asks instead for a finite reduction.

We note that the block construction of Section 10 cannot be used for this test: its drift grows like $s \log \log X$ and, summed over the eight blocks $q \leq 23$, reaches only 0.05, so neither the protected nor the unprotected version has killed anything at reachable scales. In the two simplest systems with $\nu > 0$ and divergent drift, the even moduli at residue 0 and the doubled construction of Section 10, the survivors come from a cylinder that carries no classes at all – the odd numbers – rather than from any cancellation among the classes acting there. We know of no example of divergent drift in which a positive survivor measure arises from genuine cancellation on a cylinder whose induced drift diverges. Whether such systems exist is itself open.

What Proposition 9.6 buys is not a proof but a translation: the open statement is now that a Buchstab-type ratio tends to 1 on average, for arbitrary congruence systems rather than for sieves by primes. The classical theory controls that ratio sharply in the prime case; whether it can be controlled here we do not know.

Remark 9.13. Theorem 9.3 is an equivalence, not merely a sufficient condition, and it relocates the difficulty: from controlling an infinite tail to bounding the measure of the set of *scales* at which fresh classes over-kill. By Proposition 7.9 that control cannot come from an averaged count of drift, so it must come from the arithmetic of the residues.

Heuristically, to keep w elevated on a set of scales of positive logarithmic density one needs classes of modulus comparable to X at a positive proportion of scales, hence essentially every integer as a modulus – and such a system kills everything, forcing $\nu = 0$ and removing the excursion. This is the same self-defeating tension as in Remark 9.15 and in Section 10, appearing here a third time. We have not turned it into a proof.

Remark 9.14. Theorem 9.1, by contrast, is stated as a sufficient condition only. Necessity is not established here and does not follow formally: for fixed k one has $\lim_X \big(F(X) - F_k(X)\big) = \nu_k - \nu$, which tends to 0 with k regardless, so all of the content lies in the supremum, and a transient excursion could in principle exceed the limit even when $\delta_{\log}(A)$ exists.

Remark 9.15. It is worth asking whether the hypothesis can be met at all, and the obvious attempt to defeat it fails for an instructive reason. One would try to manufacture transient excursions with a block of moduli in $[N, 2N]$. Aligned at 0 such a block is useless: the density of integers with a divisor in $[N, 2N]$ tends to 0 as $N \to \infty$, by a theorem of Erdős [11], as Besicovitch's construction requires. With residues in general position the block's balls are nearly independent, and it kills a profinite measure d_N that stays near $1 - \prod_{N \le n \le 2N}(1 - 1/n) \to \frac{1}{2}$: measured on $\widehat{\mathbb{Z}}$ with random residues, 0.516, 0.513 and 0.507 at $N = 20, 40, 80$, against 0.426, 0.413 and 0.392 for the same moduli aligned at 0. So if such a block lies beyond the head its fresh contribution is of order $d_N \nu_k$, giving

$$F(X) - F_k(X) \ \gtrsim\ d_N \nu_k \Big(1 - \frac{\log N}{\log X}\Big),$$

which at $X = N^2$ is about $d_N \nu_k / 2$, apparently independent of k. But the strategy is self-defeating. To keep this bounded away from 0 for every k one needs infinitely many blocks with d_N bounded below; if the blocks use disjoint sets of primes, so that their kill events are independent, J of them leave survivor measure $\prod(1 - d_N) \approx 2^{-J}$, so $\nu = 0$ and the conclusion of the theorem holds trivially, while blocks correlated by a common protector fall under Theorem 8.1. What is left is a sequence of blocks with d_{N_j} bounded below, correlated neither by independence nor by a common protector, and with $\nu > 0$; we have no argument excluding it, and it is a special case of the refined Question. So the natural counterexample strategy does not work as stated, but neither is it refuted, and we do not claim this settles necessity.

10 Four refuted routes, and a bound on a fifth

We record, with constructions, four natural approaches that do not work, and a bound on a fifth. These appear to us more useful to a reader attacking the problem than the positive results. The fifth is an attempt to *refute* the conjecture rather than prove it.

Divergence need not lie in a single channel

Split the primes into two sets Q and R, each of divergent reciprocal sum, say by alternating; partition R into consecutive blocks P_q, one per $q \in Q$, each the shortest block of harmonic weight $S_q := \sum_{p \in P_q} 1/p$ at least s, so that $s \le S_q < s + 1/\min P_q < s + 1$, the cutoff large enough that every block holds at least q primes; take as moduli the products qp for $p \in P_q$, list each block in increasing order and give the class of modulus qp_j the residue $j \bmod q$, so that the residue groups are the arithmetic progressions of positions modulo q. Classes from different blocks are coprime, since the block primes lie in R and the indexing primes in Q, hence always compatible; classes within a block have gcd $= q$ exactly. Hence a channel contains at most one residue group per block, and the maximal channel decomposes as $\sum_q$(heaviest group in block q). Since the weights $1/p$ decrease along a block, the heaviest group is the one containing the first prime, and for a decreasing sequence the sum of every qth term is at most $1/\min P_q + \frac{1}{q}\sum_{p\in P_q} 1/p$; dividing by q and summing,

$$\text{maximal channel} \;\le\; \sum_q \frac{1}{q\,\min P_q} \;+\; (s+1)\,P(2),$$

where $P(2) = \sum_p p^{-2}$ is the prime zeta value. Both terms converge, the first because consecutive blocks of harmonic weight at least s force, by Mertens' theorem, the least prime of the kth block to exceed $\exp(e^{ks})$ up to a constant, so that $\sum_q 1/(q \min P_q)$ is dominated by a doubly exponentially decaying series. The total drift, meanwhile, is $\sum_q S_q/q \ge \sum_q s/q$, which diverges by Mertens. (The cruder bound $s\,P(2) \approx 0.45s$ alone is not correct: the block at $q = 2$ contributes about $S_2/2 \ge s/2$ on its own.)

Bounded channels need not force death

Double every modulus in the previous construction and force even residues. Every kill set then lies in the even numbers, so every odd number survives and the survivor density is at least $1/2$, while the drift still diverges and the channels remain bounded.

The ε-split fails

The same doubled construction shows that no finite family of channels captures all but ε of a divergent drift.

Heilbronn–Rohrbach fails for shifted classes

The inequality $\delta \ge \prod(1 - 1/n_i)$ of Heilbronn and Rohrbach [18, 19], generalised by Behrend [1] and available for sets of multiples, is false for shifted systems: for moduli $2, 4, 8$ with residues $0, 1, 3$ the survivor density is 0.125 against a bound of 0.328125. Enumerating every system of three classes whose moduli are a 3-element subset of $\{2, \ldots, 12\}$ and whose residues range independently over all residues modulo those moduli gives $\sum_{\{a<b<c\}\subset\{2,\ldots,12\}} abc = 53{,}130$ systems; each is periodic modulo $\mathrm{lcm}(a, b, c)$, so its survivor density is an exact rational, and 23,034 of them – 43.4% – fall strictly below the bound, the displayed system being the extreme case. The enumeration is specified here so that it may be reproduced directly.

A bound for top-octave spikes

Since $\overline{\delta}_{\log}(A) \le \nu$ is unconditional (Theorem 3.1), a counterexample would need the killed set to exceed its own measure in upper *logarithmic* density, so F must oscillate. The natural attempt uses classes of modulus comparable to X: a class with $n \in (X/2, X]$ kills at most one element of that window, so K of them can raise the window's killed density by $2K/X$ while adding only K/X to the drift. Such spikes are real and large. Taking the class 0 mod n for every $n \in (Y/2, Y]$ and each of $Y = 2^{13}$, 2^{16}, 2^{19}, 2^{21} together, the killed density of the dyadic window $(2^k, 2^{k+1}]$

runs, for $k = 12, \dots, 20$, through (the entries 1.000 at $k = 12, 15, 18, 20$ being the four spiked windows)

$$1.000,\ 0.583,\ 0.527,\ 1.000,\ 0.692,\ 0.645,\ 1.000,\ 0.747,\ 1.000,$$

an amplitude above 0.4. Yet F does not oscillate at all: evaluated at the tops $X = 2^{k+1}$ of the same windows it rises monotonically through

$$0.077,\ 0.112,\ 0.140,\ 0.194,\ 0.222,\ 0.246,\ 0.286,\ 0.308,\ 0.341.$$

The reason is structural. A spike occupies one dyadic window, of logarithmic length $\log 2$, so its excess above the persistent level contributes $O(1/\log X)$ to the logarithmic average and vanishes. To move F one would need the elevation sustained over a constant fraction of $\log X$; but a class of modulus n kills density $1/n$ at *every* scale above n, so a sustained elevation is permanent rather than transient, accumulates across scales, and drives the survivor density to 0 – whereupon $\nu = 0$ and the logarithmic density exists trivially. We state this last step as a heuristic, not a theorem: no result here quantifies how successive excursions of intermediate moduli interact. What is proved is Proposition 9.4: isolated spikes cannot move the logarithmic average when $h = 0$.

A Verification

The numerical assertions above were checked by computation, deterministic except where a Monte Carlo value is quoted, in which case the pseudo-random seed is fixed in the program; several checks are accompanied by a control designed to fail if the property being checked were absent. The values on which the text's explicit assertions rest are reproduced by a single claims script; the Monte Carlo and window measurements quoted as illustration are reproduced by the individual programs named in the artifact, which is public at `https://github.com/ibrahimmian36/Triarius`. No theorem in this paper depends on these computations: each is proved above, and the computations serve only to exhibit the explicit examples and counterexamples, and in Remark 7.3 to show why two plausible routes fail. The programs, their recorded outputs and this paper form the public, versioned artifact at that address.

Provenance of the illustrative measurements. Each measurement quoted in a remark is produced by one named program in the artifact's `probes` directory, run without arguments: the many-protector and two-level-tower measurements of Section 8 by `verify_multiprotector.py` and `verify_tower.py`; the window ratios and the intermediate-range ratios by `verify_buchstab.py` and `verify_overkill.py`; the induced-drift measurements of the refinement remark by `verify_drift_invariance.py`; the coarsening statistics by `verify_coarsening.py`; the semiprime, protector, tree and overlapping-protector instances by `verify_quasi.py`, `verify_confined.py`, `verify_coprime_protectors.py`, `verify_tree.py` and `verify_overlap.py`; the node assignments of the chain constraint by `verify_organizable.py`; the saturation reduction by `verify_saturation.py`; the spike inequalities and the small-window excursions by `verify_spike_bound.py` and `verify_excursion2.py`; the protector test and the tower dichotomy of Section 9 by `verify_protector2.py` and `verify_infinite_tower.py`; the window ratios and the window occurrence counts of Remark 9.7 by `verify_narrow.py` and `verify_ratio.py`; the Heilbronn–Rohrbach enumeration by `verify_paper.py`; the spike display of Section 10 by the claims script; and the catalogue audit of the eleven exhibited systems by `verify_coverage.py`. The values that enter hypotheses or explicit assertions are re-derived by `tools/verify_paper_claims.py` in one run, whose output is archived with the artifact. The disjointness bound of Lemma 5.2 was confirmed against an exhaustive branch-and-bound maximum over all 104 classes of modulus at most 14, returning exactly 1.000000. The independence identity behind Theorem 5.4 was checked against a direct sieve, and a non-coprime control deviates from it. The cylinder reduction of Theorem 7.2 was verified exactly on all non-empty cylinder-class pairs for a system with misaligned parts $3, 5, 7$ and $M = 105$. The quantities in the proof of Proposition 9.11 – the reciprocal sums S_p, the measures $\mu(\widehat{K}_p)$ and $\mu(\widehat{K}_p \cap \widehat{K}_{p'})$ by inclusion–exclusion over the independent prime coordinates, and the induced drift and survivor measure of every cylinder – were computed exactly for $p \leq 150$, with the sparse-p restriction as the control on which the theorem's hypothesis fails. For Proposition 9.12 the induced survivor measure of every multi-class cylinder, the drift, and the misalignment were computed exactly on the six-group truncation, with the divergent-protector replacement as the control. The node assignments of Proposition 8.14 were searched exhaustively for four truncated systems, confirming that a depth-two slice of the tree and the multiples of 6 are coprime-layered and that the protector and semiprime systems are not. For Theorem 8.12 an explicit tree of depth four – children k^2 of $k^2 + 1$ residues at depth k, attached primes from pools disjoint by depth and coprime to $2, 5, 17$, nodes of large drift in proportion $1/k^2$ – has 617 nodes and 23,361 classes; its survivor measure 0.96432, computed as the product $\prod_q (1 - k_q/q)$ from the proof of Lemma 8.10, agrees with a Monte Carlo draw of 20,000 points (0.9646); at each of four levels N the first crossings of Lemma 8.11 are pairwise disjoint with total measure equal to $\mu\{g > N\}$ exactly, and the light drift is below N; and the window survivor density agrees with the visible profinite measure to three decimals at $X = 10^4, 10^5, 10^6$ and to within 10^{-3} at 4×10^6, which illustrates the theorem and proves nothing. Finally, as a safeguard against misattribution, the catalogue of systems exhibited in this paper – eleven of them, from the aligned residues of Theorem 5.9 to the pairwise products of Theorem 8.18 – was audited mechanically: for each, the finitely checkable hypotheses of the theorem the text cites for it are verified on a truncation, the infinitary ones – divergence, distinctness, escape – being established in the text where each system is introduced; the finite checks pass. The saturation reduction of Theorem 8.4 was checked exactly on the finite systems described there.

Declaration of generative AI use. Large language models – Anthropic's Claude (most recently Claude Fable 5.1) and OpenAI's GPT-5.6 – were used throughout this work: to explore and draft arguments, to draft and revise the text, to write the programs in the artifact, and, in separate sessions without access to earlier reviews, to review the manuscript for errors. The authors directed the work, checked every proof, and take full responsibility for the content. Every number quoted in the paper is reproduced by a named program of the artifact, whose recorded outputs it contains.

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